

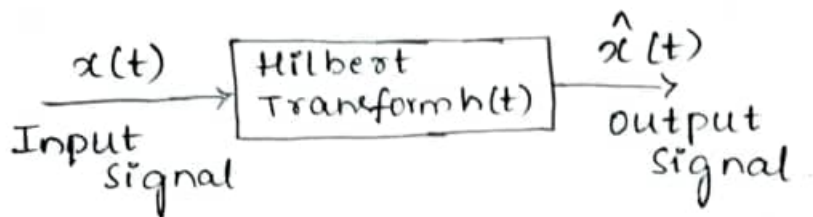
DIGITAL COMMUNICATION (18EC61)

MODULE - 1

BANDPASS SIGNAL TO EQUIVALENT LOW PASS

* HILBERT TRANSFORM

- Fourier, Laplace and Z-transform involves the change from Time-Domain representation of a signal to the Frequency-domain representation of the signal.
- The resulting signals are equivalent representation of the same signal in terms of Frequency, S-domain and Z-domain.
- In Contrast, The Hilbert transform does not involve change of domain, Unlike many other transforms.
- Hilbert Transform is not an equivalent representation of the signal, rather it is entirely a different signal but in the same Domain.
- Fourier Transform is Useful for evaluating the frequency content of a signal. It provides mathematical basis for analyzing and designing Frequency - selective filters for the separation of signals on the basis of their frequency content.
- Hilbert Transform is a method of separating signals based on Phase selectivity, which uses phase shift between pertinent signals.
- When the phase angles of all the components of a given signal are shifted by $\pm 90^\circ$, the resulting function of time is known as Hilbert transform of the signal.
- It is also called as Quadrature filter as it provides $\pm 90^\circ$ phase shift.
- Hilbert transformer is a $\pm 90^\circ$ phase shifter in time domain without changing Amplitude value.



- Hilbert transform of a signal $x(t)$ is denoted by $\hat{x}(t)$
- Let $x(t)$ be a fourier transformable signal with its fourier transform $X(f)$.
- The Hilbert Transform $\hat{x}(t)$ of a input signal $x(t)$ be defined as $\hat{x}(t) = x(t) * h(t)$

$$\text{ie, } \boxed{\hat{x}(t) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{x(\tau)}{(t-\tau)} \cdot d\tau} \quad \text{--- (1)}$$

- Hilbert Transform is a linear function. So, its Inverse Hilbert transform is given by ,

$$\boxed{x(t) = \frac{-1}{\pi} \int_{-\infty}^{\infty} \frac{\hat{x}(\tau)}{(t-\tau)} d\tau} \quad \text{--- (2)}$$

- From definition, Hilbert Transform $\hat{x}(t)$ is the Convolution of $x(t)$ with the time function $h(t) = \frac{1}{\pi t}$.

$$\therefore \hat{x}(t) = x(t) * \frac{1}{\pi t} \quad \text{--- (3)}$$

WKT. Convolution in time domain is equal to product in Freq Domain.

Applying FT to equation (3)

$$\text{FT}\{x(t)\} = X(f) \text{ \& \text{ FT}\{\hat{x}(t)\} = \hat{X}(f)}$$

$$\text{FT}\left\{\frac{1}{\pi t}\right\} = -j \text{sgn}(f)$$

WKT,

$$\text{Sgn}(t) \xleftrightarrow{\text{FT}} \frac{2}{j\omega} = \frac{1}{j\pi f}$$

using Duality property,

$$\boxed{\frac{1}{\pi t} \xleftrightarrow{\text{FT}} -j \text{sgn}(f)}$$

Studied smart, not hard — thanks to VTUSync.in

Note

$\because \omega = 2\pi f$
 * Duality property
 If $g(t) \rightleftharpoons G(f)$
 then $G(t) \rightleftharpoons g(-f)$
 * $\text{sgn}(-f) = -\text{sgn}(f)$

Where $\text{sgn}(f)$ is the signum function, defined in frequency domain as,

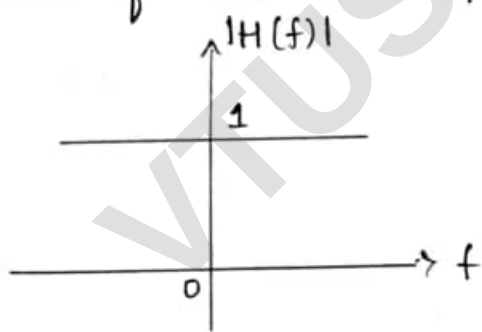
$$\text{sgn}(f) = \begin{cases} 1 & ; f > 0 \\ 0 & ; f = 0 \\ -1 & ; f < 0 \end{cases}$$

$\therefore$ Fourier transform of Eq(3) is given by

$$\hat{X}(f) = -j \text{sgn}(f) \cdot X(f) \quad \text{--- (4)}$$

→ Equation (4) states that the given Fourier transformable signal $x(t)$. We may obtain the Fourier Transform of its Hilbert transform $\hat{x}(t)$ by passing $x(t)$ through LTI system whose frequency response is equal to $-j \text{sgn}(f)$.

→ If we multiply +ve spectrum of $x(t)$ [+ve freq of $x(f)$] by $-j$ and -ve spectrum of $x(t)$ [ie, -ve freq of $x(f)$] by $+j$, we obtain the Hilbert transform of $x(t)$ in Frequency Domain [ie, $\hat{X}(f)$].



(a) —

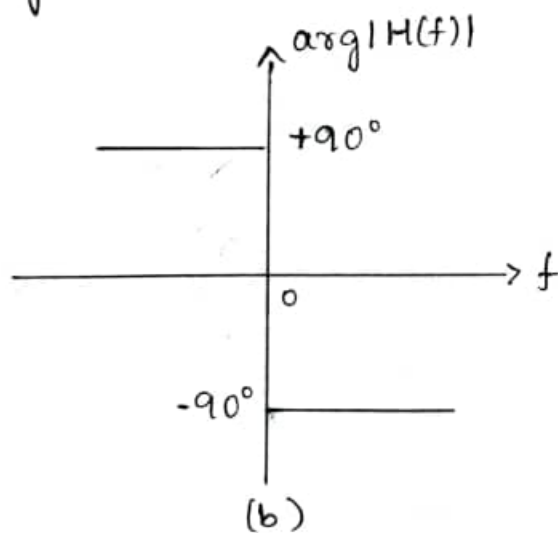


fig: a] Magnitude response

b] phase response of Hilbert Transform.

Applications

- In the Generation of SSB modulation.
- Representation of Bandpass signal mathematically.

* Hilbert Transform of Basic Signals.

	Time Function	Hilbert Transform
1)	$m(t) \cos(2\pi f_c t)$	$m(t) \sin(2\pi f_c t)$
2)	$m(t) \sin(2\pi f_c t)$	$-m(t) \cos(2\pi f_c t)$
3)	$\cos(2\pi f_c t)$	$\sin(2\pi f_c t)$
4)	$\sin(2\pi f_c t)$	$-\cos(2\pi f_c t)$
5)	$\frac{\sin t}{t}$	$\frac{1 - \cos t}{t}$
6)	$\text{rect}(t)$	$-\frac{1}{\pi} \ln \left[\frac{t - 1/2}{t + 1/2} \right]$
7)	$\delta(t)$	$1/\pi t$
8)	$\frac{1}{1+t^2}$	$\frac{1}{1+t^2}$
9)	$1/t$	$-\pi \delta(t)$

* Properties of Hilbert Transform

Property 1: A signal $x(t)$ and its Hilbert transform $\hat{x}(t)$ have the same magnitude spectrum.

$$|X(f)| = |\hat{X}(f)|$$

Proof: The signal $x(t)$ be Fourier transformable with Fourier transform $X(f)$.

Let $\hat{x}(t)$ be its Hilbert transform & $\hat{X}(f)$ be the Fourier transform of $\hat{x}(t)$. Then

$$\hat{X}(f) = -j \text{sgn}(f) \cdot X(f)$$

$$|\hat{X}(f)| = |-j \text{sgn}(f) \cdot X(f)|$$

$$|\hat{X}(f)| = |X(f)|$$

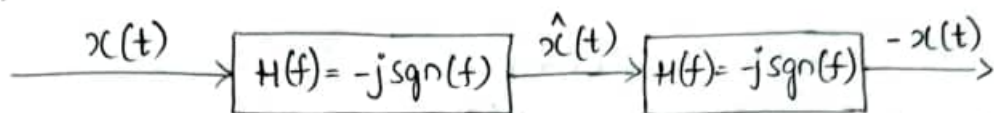
$$\because |-j \text{sgn}(f)| = 1$$

Hence proved

Property 2: If $\hat{x}(t)$ is the Hilbert transform of $x(t)$ then the Hilbert transform of $\hat{x}(t)$ is $-x(t)$.

$$\boxed{\arg[x(f)] = -\arg\{\hat{G}(f)\}}$$

Proof:



Cascading two ideal two-port devices to obtain double Hilbert Transform.

$$\hat{x}(t) = x(t) * h(t)$$

Taking FT of above equation,

$$\hat{X}(f) = X(f) \cdot H(f) \quad [\because H(f) = -j \operatorname{sgn}(f)]$$

$$\hat{X}(f) = -j \operatorname{sgn}(f) \cdot X(f)$$

Hilbert Transform of $\hat{x}(t)$ is given by $\hat{\hat{x}}(t)$ — (1)

ie, $\hat{\hat{x}}(t) = \hat{x}(t) * h(t)$

Taking Fourier transform we get,

$$\hat{\hat{X}}(f) = \hat{X}(f) \cdot H(f)$$

$$\hat{\hat{X}}(f) = [-j \operatorname{sgn}(f) \cdot X(f)] [-j \operatorname{sgn}(f)]$$

WKT, $\operatorname{sgn}(f) \cdot \operatorname{sgn}(f) = 1$ and $j^2 = -1$.

$$\therefore \hat{\hat{X}}(f) = -X(f)$$

Taking IFT of Eq(2) we get, Hilbert transform of $\hat{x}(t)$ — (2)

ie, $\boxed{\hat{\hat{x}}(t) = -x(t)}$ — (3)

Hence proved.

Property 3: A signal $x(t)$ and its Hilbert transform $\hat{x}(t)$ are orthogonal over the entire time interval $(-\infty, \infty)$

$$\boxed{\int_{-\infty}^{\infty} x(t) \cdot \hat{x}(t) dt = 0}$$

Proof: To prove that $x(t)$ and $\hat{x}(t)$ are orthogonal, we have to prove that

$$\text{i.e., } \int_{-\infty}^{\infty} x(t) \cdot \hat{x}(t) dt = 0$$

Proof: Using Parseval's theorem, $E_{TD} = E_{FD}$.

$$\text{i.e., } \int_{-\infty}^{\infty} x(t) \cdot \hat{x}(t) dt = \int_{-\infty}^{\infty} X(f) \cdot \hat{X}^*(f) df \quad \text{--- (1)}$$

WKT, for a real valued signal,

$$\hat{X}^*(f) = \hat{X}(-f)$$

Rewriting Eq(1) we get

$$\int_{-\infty}^{\infty} x(t) \cdot \hat{x}(t) dt = \int_{-\infty}^{\infty} X(f) \cdot \hat{X}(-f) df \quad \text{--- (2)}$$

$$\text{WKT, } \hat{X}(f) = -j \operatorname{sgn}(f) \cdot X(f)$$

$$\hat{X}(-f) = -j \operatorname{sgn}(-f) \cdot X(-f) = j \operatorname{sgn}(f) \cdot X(-f)$$

Substituting the value of $\hat{X}(-f)$ in Eq(2) we get

$$\begin{aligned} \int_{-\infty}^{\infty} x(t) \hat{x}(t) dt &= \int_{-\infty}^{\infty} X(f) \cdot j \operatorname{sgn}(f) X(-f) df \\ &= j \int_{-\infty}^{\infty} X(f) \cdot X(-f) \operatorname{sgn}(f) df \\ &= j \int_{-\infty}^{\infty} |X(f)|^2 \cdot \operatorname{sgn}(f) df \end{aligned}$$

$$\therefore \int_{-\infty}^{\infty} x(t) \cdot \hat{x}(t) dt = 0$$

WKT, $x(f) \cdot x(-f) = |X(f)|^2$ which is even function & $\operatorname{sgn}(f)$ is odd function. Hence Integration of odd function over the limits $-\infty$ to ∞ is zero. Hence proved.

Additional Properties

1. The Hilbert Transform of a real signal is also real.
2. The Hilbert Transform of an even function is odd and odd function is even.
3. The magnitude spectra of a signal $x(t)$ & its HT $\hat{x}(t)$ are identical.
4. If $\hat{x}(t)$ is the HT of $x(t)$ then HT of $\hat{x}(t)$ is $-x(t)$.

* Hilbert Transform of Low Pass Signal

The below figure depicts Fourier transform of a low pass signal $x(t)$, whose frequency content extends from $-\omega$ to $+\omega$

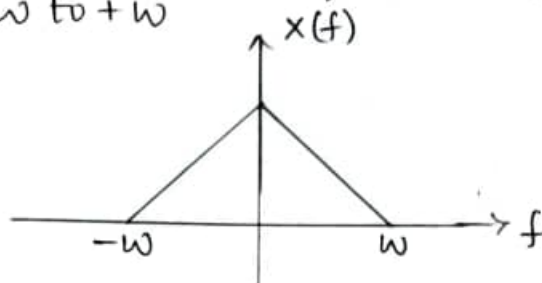


fig: Spectrum of signal $x(t)$

Applying Hilbert Transform to this signal yields a new signal $\hat{x}(t)$ whose Fourier transform ($\hat{x}(f)$) is shown as follows

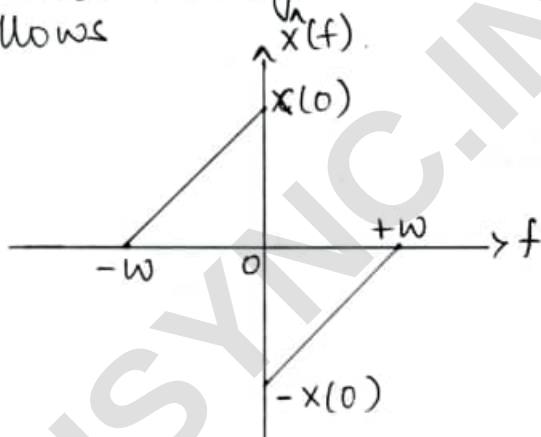


fig: Spectrum of Hilbert transform of $\hat{x}(t)$

* PRE-ENVELOPES

→ The Hilbert Transform of a signal is defined for both positive and negative frequencies.

→ We can modify the frequency content of a real valued signal such that all negative or positive frequency components are completely eliminated. With the aid of Pre-Envelope concept

→ The pre-envelope of a signal is a complex valued signal with either only positive frequencies or only negative frequencies. And denoted as $x_+(t)$ & $x_-(t)$

→ It is a complex signal generated by adding a signal with its Hilbert transform in Quadrature.

The Complex valued signal pre-envelope is formally defined as

$$\boxed{x_+(t) = x(t) + j\hat{x}(t)} \quad (1)$$

where $x_+(t)$ — positive pre-envelope of $x(t)$

$\hat{x}(t)$ — Hilbert Transform of $x(t)$

$x(t)$ — a real valued signal.

Applying Fourier Transform for eq(1) we get

$$\begin{aligned} X_+(f) &= X(f) + j[-j \operatorname{sgn}(f) \cdot X(f)] \\ &= X(f) + \operatorname{sgn}(f) \cdot X(f) \quad (\because j^2 = -1) \\ X_+(f) &= X(f) [1 + \operatorname{sgn}(f)] \end{aligned} \quad (2)$$

We know that

$$\operatorname{sgn}(f) = \begin{cases} 1; & f > 0 \\ 0; & f = 0 \\ -1; & f < 0. \end{cases}$$

Substituting the values of $\operatorname{sgn}(f)$ in eq(2) we get,

$$X_+(f) = \begin{cases} 2X(f) & ; f > 0 \\ X(0) & ; f = 0 \\ 0 & ; f < 0 \end{cases} \quad (3)$$

where $X(0)$ is the value of $X(f)$ at the origin $f=0$.

Equation (3) clearly shows that the pre-envelope of the signal $x(t)$ has no frequency content for all negative frequencies.

Hence pre-Envelope for positive frequencies will not contain any negative frequencies.

We may determine pre-envelope for positive frequencies i.e., $x_+(t)$ in one of two equivalent procedures.

1. Time-Domain Procedure: Given the signal $x(t)$, we use Eq(1) to compute pre-envelope $x_+(t)$.

2. Frequency-Domain Procedure

→ Firstly determine Fourier transform of the signal $x(t)$. i.e., $X(f)$

→ Then use Eq(3) to find pre-envelope of positive frequencies in frequency domain i.e., $X_+(f)$

→ Then obtain inverse Fourier transform of $X_+(f)$ to obtain

$$x_+(t) = 2 \int_0^{\infty} X(f) \exp(j2\pi ft) \cdot df$$

Depending on the description of the signal we may select any of the procedure to obtain pre-envelope for positive frequencies.

Pre-envelope for Negative Frequencies.

Consider a real valued Fourier transformable signal $x(t)$. Such that $X(f)$ be its Fourier transform.

The Pre-envelope for negative frequencies is defined as

$$x_-(t) = x(t) - j \hat{x}(t) \quad (4)$$

The two pre-envelopes $x_+(t)$ & $x_-(t)$ are simply complex conjugate of each other.

$$\text{i.e., } x_-(t) = x_+^*(t)$$

Taking Fourier transform of Eq(4), we get

$$\begin{aligned} X_-(f) &= X(f) - j \hat{X}(f) \\ &= X(f) - j[-j \operatorname{sgn}(f) X(f)] \\ &= X(f) - \operatorname{sgn}(f) X(f) \end{aligned}$$

$$X_-(f) = X(f) [1 - \operatorname{sgn}(f)] \quad (5)$$

Applying the values of $\operatorname{sgn}(f)$ to Eq(5) we get Pre-envelope for negative frequencies as.

$$X_-(f) = \begin{cases} 0 & ; f > 0 \\ X(f) & ; f = 0 \\ 2X(f) & ; f < 0 \end{cases} \quad (6)$$

From eq(6), it is clear that Pre-envelope for negative frequencies donot contain any positive frequencies of the spectrum.

→ The sum of $x_+(t)$ and $x_-(t)$ is exactly twice the original signal $x(t)$.

$$\text{ie, } x_+(t) + x_-(t) = [x(t) + j\hat{x}(t)] + [x(t) - j\hat{x}(t)]$$

$$x_+(t) + x_-(t) = 2x(t)$$

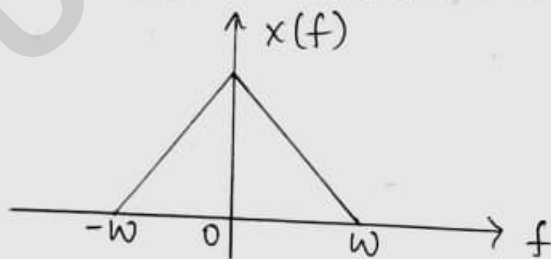
$$x(t) = \frac{x_+(t) + x_-(t)}{2}$$

→ The Spectrum of pre-envelope $x_+(t)$ is nonzero only for positive frequencies, hence the use of '+' as the subscript.

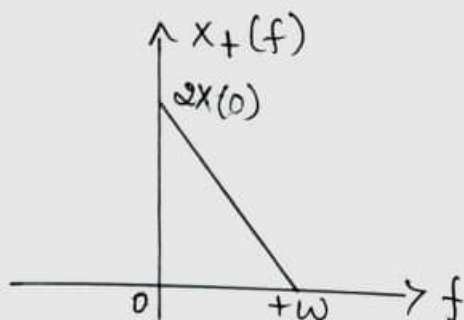
→ The Spectrum of pre-envelope $x_-(t)$ is non zero only for negative frequencies. hence the use of '-' as Subscript.

Pre-envelope of a Low Pass signal

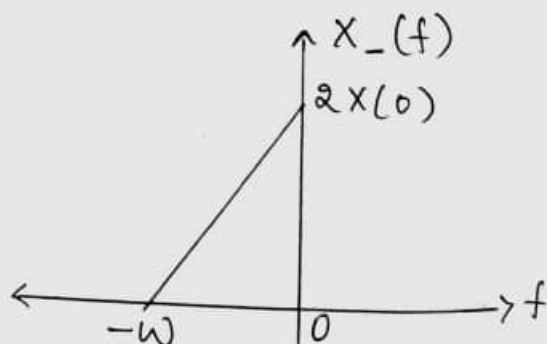
Consider a low pass signal $x(t)$ with its fourier transform $x(f)$ as shown below.



Spectrum of Pre-envelope $x_+(t)$ and $x_-(t)$ are shown below



Spectrum of $x_+(t)$
($0 < f < w$)



Spectrum of $x_-(t)$
($w < f < 0$)

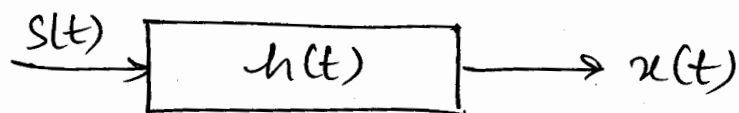
PRACTICAL IMPORTANCE OF HILBERT TRANSFORMATION.

- The pre-envelope simplifies the analysis of band pass signals and band pass systems in signal theory.
- By applying the concept of pre-envelope to a bandpass signal, the signal is transformed into an equivalent low-pass representation.
- In a corresponding way, a bandpass filter is transformed into its own equivalent low-pass representation.
- Both transformations, rooted in the Hilbert Transform, play a key role in the formulation of modulated signals and their demodulation.

COMPLEX ENVELOPES OF BAND PASS SIGNALS.

- Pre-envelope applies to any real valued signal - low pass or band pass, only requirement is that the signal be power transformable.
- we will restrict attention to bandpass signals / system because it is directly applicable to analog modulation and digital modulation schemes.

Henceforth we will use $s(t)$ to denote a modulated signal. When such a signal is applied to the input of a Band-pass system, such as a communication channel, we will use $x(t)$ to denote the resulting system output (channel output). And as before we will use $h(t)$ as the impulse response of the system.



Bandpass s/m & commⁿ channel

$s(t) \rightarrow$ Modulated signal

$x(t) \rightarrow$ channel op.

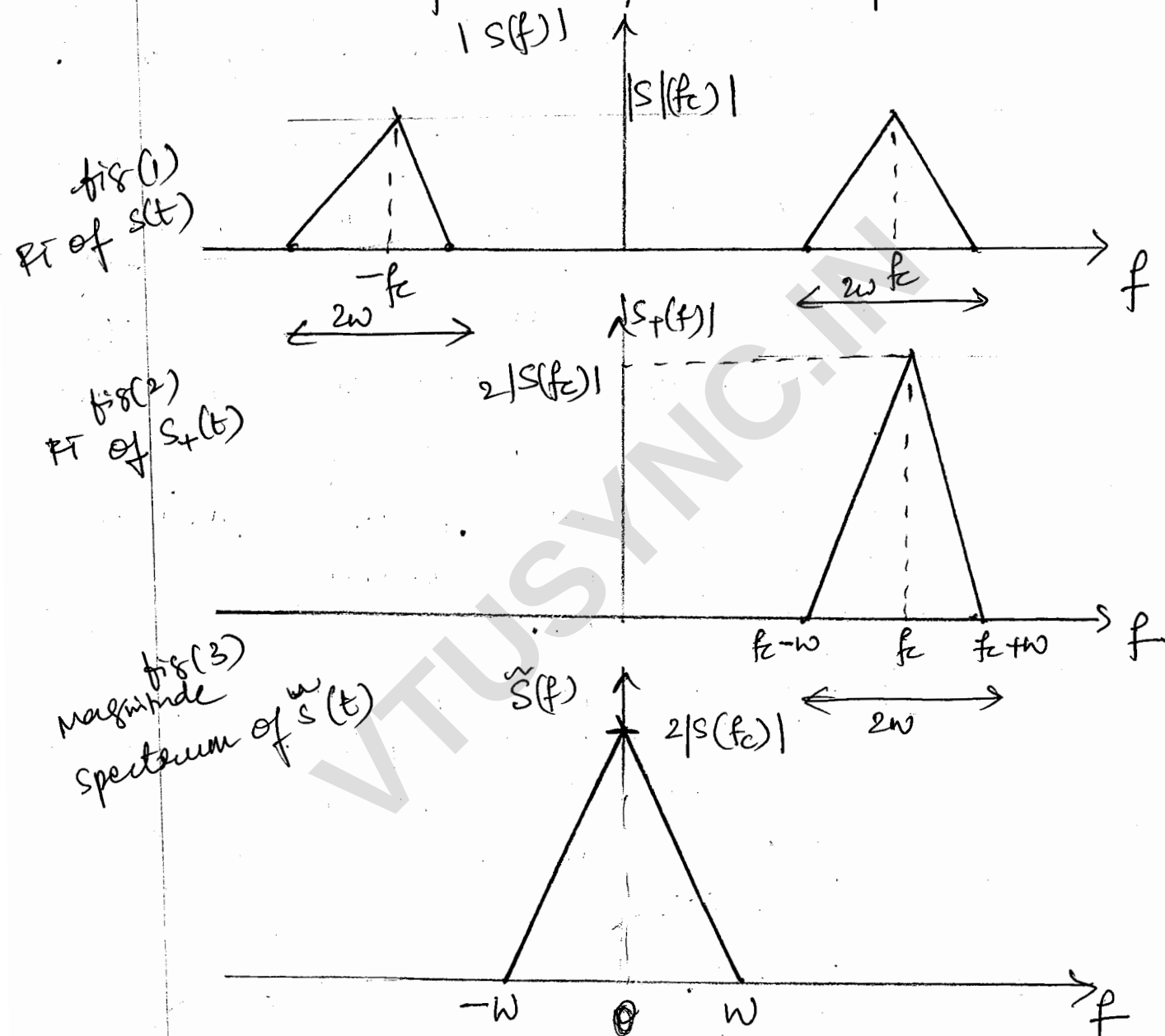
- Let the bandpass signal $s(t)$ and its Fourier Transform is represented by $S(f)$.
- Assume that FT $S(f)$ is essentially confined to a band of frequencies of total extent $2W$, center about some frequency $\pm f_c$ as shown in fig (1). where $f_c \rightarrow$ carrier frequency.
- Bandwidth $2W$ is small compared with f_c . So we may refer $s(t)$ as a narrow band signal.
- The terms narrowband and bandpass are used interchangeably.

Let the Pre-envelope of the narrow band signal $s(t)$ be expressed in the form

$$s_+(t) = \tilde{s}(t) \exp(j2\pi f_c t) \rightarrow (12)$$

where $\tilde{s}(t)$ = complex envelope of band pass signal $s(t)$.

Eqn (12) may be viewed as the basis of a definition for the complex envelope $\tilde{s}(t)$ in terms of the pre-envelope $s_+(t)$.



- We find that the Spectrum of the pre-envelope $S_+(t)$ is limited to the positive frequency band $(f_c - W) \leq f \leq (f_c + W)$ as shown in fig (2).
- Applying the frequency Shifting property of the Fourier Transform to eqn (12) we find that the spectrum of the complex envelope $\tilde{S}(t)$ is correspondingly limited to the band $(-W \leq f \leq W)$ and centered at the origin $f=0$ as shown in fig (3).
- The complex envelope $\tilde{S}(t)$ of the bandpass signal $S(t)$ is a complex low pass signal.
- Mapping from Bandpass signal $S(t)$ to the complex low pass signal $\tilde{S}(t)$ is summarised as follows.
 - (a) Information content of the modulated signal $S(t)$ is fully preserved in the complex envelope $\tilde{S}(t)$.
 - (b) Analysis of the band pass signal $S(t)$ is complicated by the presence of f_c . In contrast the complex envelope $\tilde{S}(t)$ dispenses with f_c making its analysis simpler to deal with.
 - (c) The use of $\tilde{S}(t)$ requires having to handle complex notation.

CANONICAL REPRESENTATION OF BAND PASS SIGNALS

- By definition, the real part of the pre-envelope $S_+(t)$ is equal to the original bandpass signal $s(t)$.
- We may therefore express the bandpass signal $s(t)$ in terms of its corresponding complex envelope $\tilde{s}(t)$ as

$$s(t) = \text{Re} [\tilde{s}(t) \exp(j2\pi f_c t)] \rightarrow (13)$$

where $\text{Re}[\cdot]$ denotes the real part of the quantity

- Since in general $\tilde{s}(t)$ is the complex valued quantity, we can express $\tilde{s}(t)$ in the Cartesian form as

$$\tilde{s}(t) = S_I(t) + jS_Q(t) \rightarrow (14)$$

where $S_I(t)$ & $S_Q(t)$ are both real valued low pass functions.

- We may therefore use eqn (13) & (14) to express the original band-pass signal $s(t)$ in the Canonical & Standard form

$$\begin{aligned} s(t) &= \text{Re} [(S_I(t) + jS_Q(t)) \exp(j2\pi f_c t)] \\ &= \text{Re} [S_I(t) e^{j2\pi f_c t} + jS_Q(t) e^{j2\pi f_c t}] \\ &= \text{Re} [S_I(t) + jS_Q(t)] e^{j2\pi f_c t} \end{aligned}$$

$$\begin{aligned}
 s(t) &= \operatorname{Re} [S_I(t) + j S_Q(t)] [\cos 2\pi f_c t + j \sin 2\pi f_c t] \\
 &= \operatorname{Re} \left\{ \underline{S_I(t) \cos 2\pi f_c t} + j S_I(t) \cos 2\pi f_c t + j S_Q(t) \sin 2\pi f_c t \right. \\
 &\quad \left. - \underline{S_Q(t) \sin 2\pi f_c t} \right\}
 \end{aligned}$$

$$\boxed{S(t) = S_I(t) \cos 2\pi f_c t - S_Q(t) \sin 2\pi f_c t} \longrightarrow \text{PASSBAND SIGNAL WAVE FORM} \quad (13)$$

[$f_c \gg W$]

where $S_I(t) \rightarrow$ In phase component of the bandpass signal $S(t)$

$S_Q(t) \rightarrow$ quadrature phase component or quadrature component of signal $S(t)$.

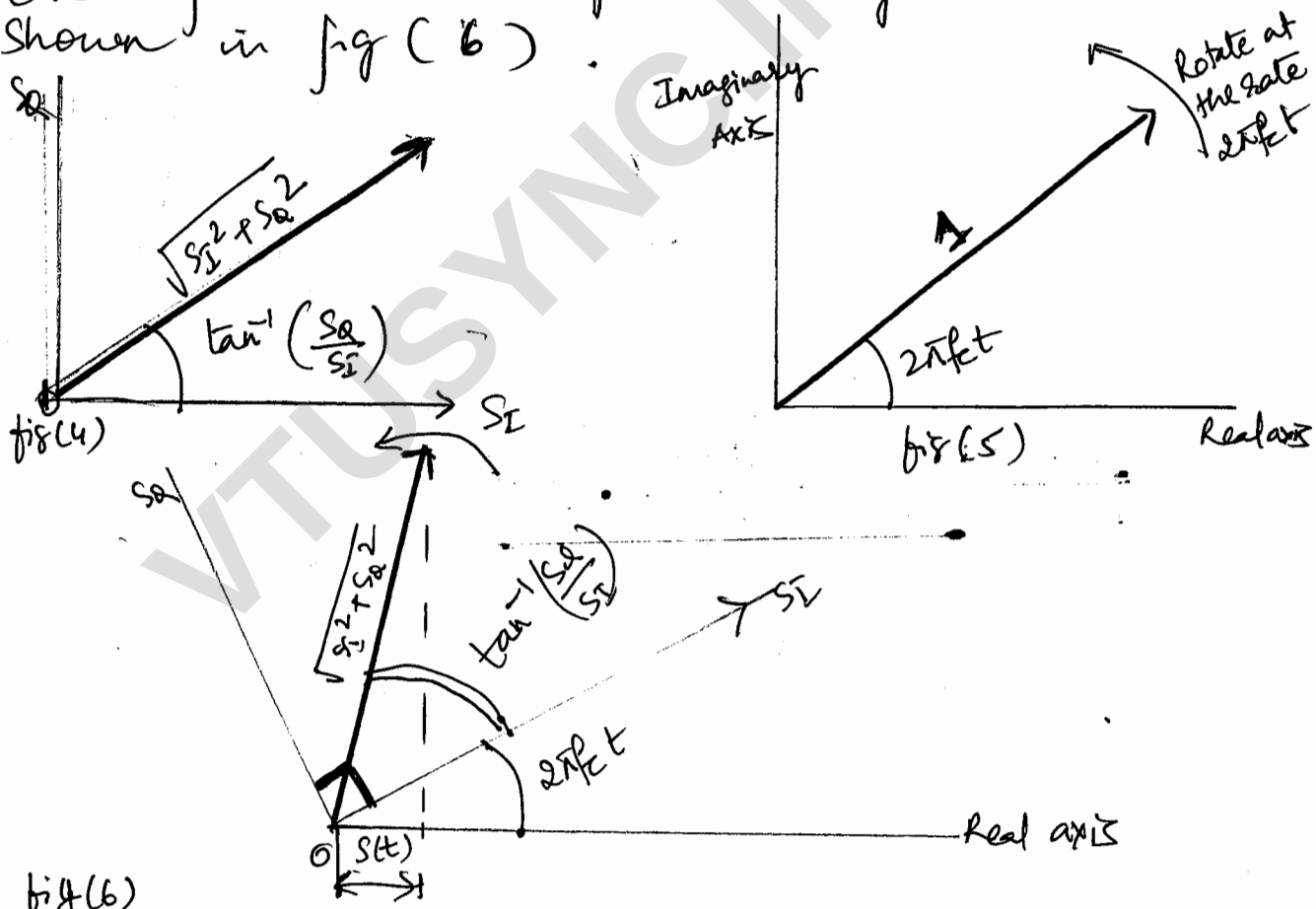
- If $\cos 2\pi f_c t * S_I(t) \rightarrow$ reference sinusoidal carrier
 $\sin 2\pi f_c t * S_Q(t) \rightarrow$ is in phase quadrature with respect to $(\cos 2\pi f_c t)$.

- According to eqn (13), the complex envelope $\tilde{S}(t)$ may be pictured as time varying phasor positioned at the origin of the (S_I, S_Q) -plane as shown in fig (4).

- Fig (5) represents the phasor representation of the complex exponential $e^{j 2\pi f_c t}$.

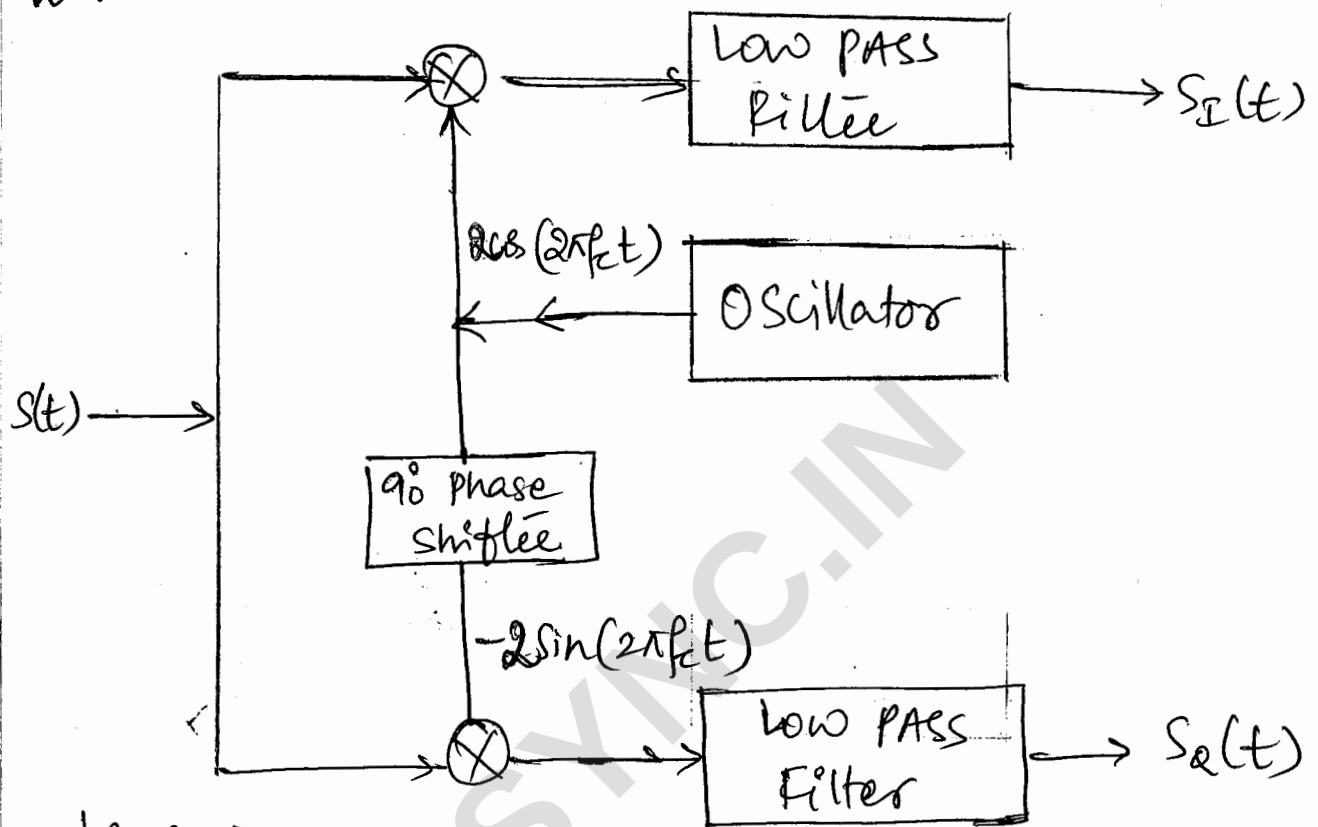
In the eqn (13), the complex envelope $\tilde{S}(t)$ is multiplied by the complex exponential $\exp(j2\pi f_c t)$. The angles of these 2 phasors add and their lengths multiply as shown in fig (6).

- We show that (S_I, S_Q) - phase rotating with an angular velocity equal to $2\pi f_c$ radians per second.
- The complex envelope $\tilde{S}(t)$ moves in (S_I, S_Q) plane and at the same time the plane itself rotates about the origin.
- The original bandpass signal $S(t)$ is the projection of this time varying phasor on a fixed line representing the real axis as shown in fig (6).



Deriving $S_I(t)$ & $S_Q(t)$ from $S(t)$ ANALYSER

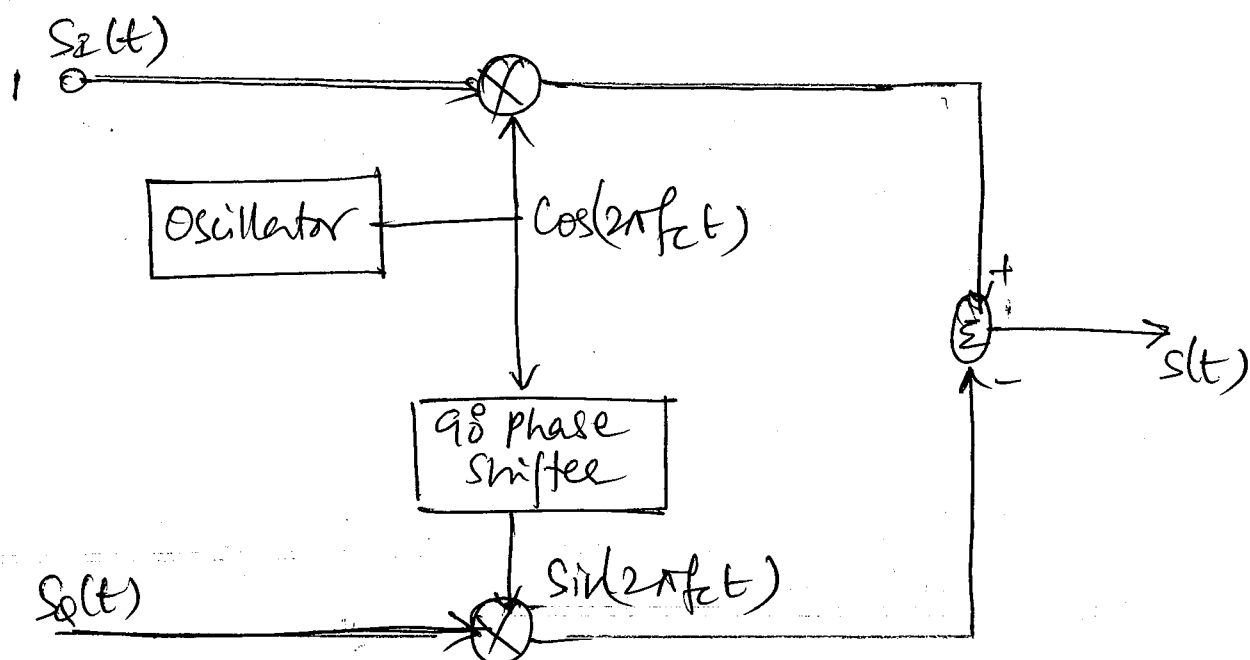
- $S_I(t)$ and $S_Q(t)$ are low pass signals limited to the band $-\omega \leq f \leq \omega$, may be extracted from the band pass signal $S(t)$ using the scheme as shown in fig (7).
- Both low pass filters are designed identically each with a Bandwidth equal to ω .



fig(7) : Analyser.

Reconstructing $S(t)$ from $S_I(t)$ & $S_Q(t)$ [SYNTHESIZER]

- To reconstruct $S(t)$ from its in phase and quadrature components, we may use the scheme as shown in fig (8).



fig(8): SYNTHESIZER

- The mapping from $S_I(t)$ and $S_Q(t)$ combined into $S(t)$ is known as passband modulation.

POLAR REPRESENTATION OF BAND PASS SIGNALS

W.K.T. Cartesian form of the complex envelope $\tilde{S}(t)$ of the bandpass signal $S(t)$ is given by $\tilde{S}(t) = S_I(t) + j S_Q(t) \rightarrow (16)$

we may write/define $\tilde{S}(t)$ in the polar form as

$$\tilde{S}(t) = a(t) \exp(j\phi(t)) \rightarrow (17)$$

where $a(t) \rightarrow$ Natural envelope of bandpass signal $S(t)$.

$\phi(t) \rightarrow$ phase of the signal.

- Both $a(t)$ & $\phi(t)$ are real valued low pass functions.

- Based on the polar representation, the original band pass signal $s(t)$ is itself defined by

$$\tilde{s}(t) = a(t) \cos[2\pi f_c t + \phi(t)] \rightarrow (18)$$

RELATION BETWEEN CARTESIAN & POLAR REPRESENTATION OF BAND PASS SIGNAL.

- The envelope $a(t)$ and phase $\phi(t)$ of a band pass signal $s(t)$ are respectively related to inphase and quadrature components $s_I(t)$ & $s_Q(t)$ as follows.

$$\begin{aligned} a(t) &= \sqrt{s_I^2(t) + s_Q^2(t)} \\ \phi(t) &= \tan^{-1} \left(\frac{s_I(t)}{s_Q(t)} \right) \end{aligned} \rightarrow (19)$$

Conversely

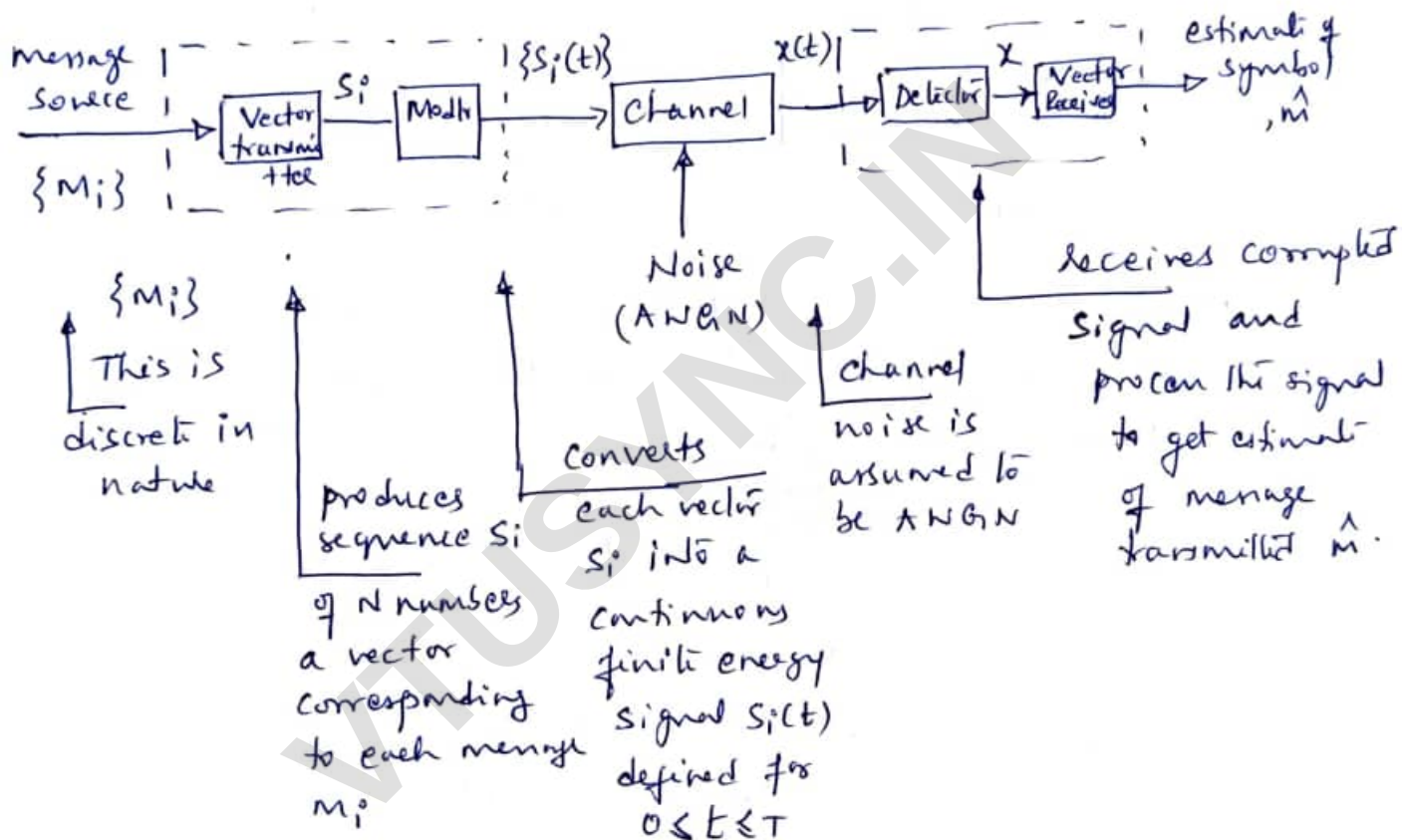
$$\begin{aligned} s_I(t) &= a(t) \cos[\phi(t)] \\ s_Q(t) &= a(t) \sin[\phi(t)] \end{aligned} \rightarrow (20)$$

Signalling over AWGN channels.

Geometric representation of signals.

Text 1: 7.1, 7.2, 7.3, 7.4

A typical Digital communication system.



Let us consider a message source with alphabet size M .
Let $m_1, m_2, m_3, \dots, m_M$ represents message symbols.
Let M symbols are equiprobable in nature, and source emits one of M symbols every T seconds.

∴ prior probabilities $P(M_i)$ is

$$P(M_i) = \frac{1}{M} \quad i = 1, 2, \dots, M \quad \text{--- (1)}$$

output of message source presented to vector source which produces a vector of real numbers

$$S_i = \begin{bmatrix} s_{i1} \\ s_{i2} \\ \vdots \\ s_{iN} \end{bmatrix} \quad i = 1, 2, \dots, M \quad \text{--- (2)}$$

where N is dimension of vector & $N \leq M$.

The input to the modulator is this S_i (eqn (2)). The modulator constructs a distinct finite energy signal $s_i(t)$ of duration T seconds.

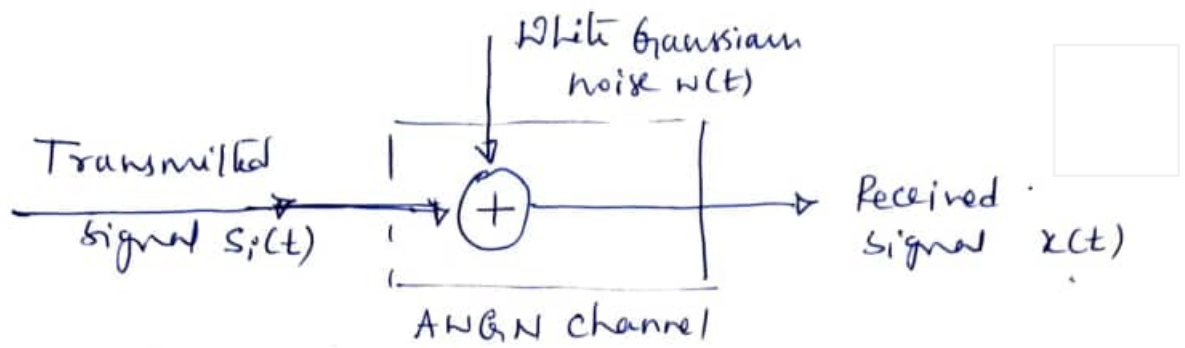
$$E_i = \int_0^T s_i^2(t) dt, \quad i = 1, 2, \dots, M \quad \text{--- (3)}$$

every message source output $M_i, i = 1, 2, \dots, M$, a distinct finite energy signal $s_i(t)$ of duration T seconds, transmitted through the channel.

The channel is assumed to have sufficient BW, signal $s_i(t)$ is corrupted by noise, this noise is additive, zero mean, stationary white Gaussian noise process $w(t)$

∴ the received signal, $x(t)$ is

$$x(t) = s_i(t) + w(t) \quad \text{--- (4)}$$

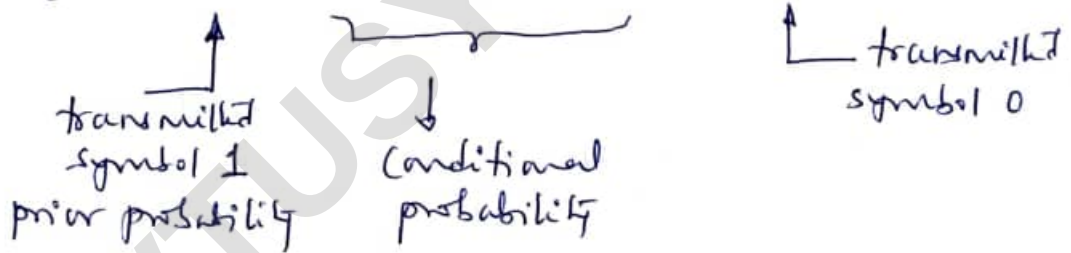


model of AWGN channel

however due to the presence of noise, receiver will inevitably make occasional errors.

hence Probability of symbol error P_e is

$$P_e = P(m_i) P(\hat{m} = 0 | 1 \text{ sent}) + P(m_i) P(\hat{m} = 1 | 0 \text{ sent})$$



Geometric Representation of Signals.

The importance of geometric representation of signals is to represent any set of M energy signals $\{s_i(t)\}$ as linear combinations of N orthonormal basis functions, where $N \leq M$.

$$\therefore s_i(t) = \sum_{j=1}^N s_{ij} \phi_j(t), \quad \begin{cases} 0 \leq t \leq T & \leftarrow \text{Symbol duration} \\ i = 1, 2, \dots, M \end{cases} \quad \text{--- (1)}$$

$$\text{Where } s_{ij} = \int_0^T s_i(t) \phi_j(t) dt \quad \begin{cases} i = 1, 2, \dots, M \\ j = 1, 2, \dots, N \end{cases}, N \leq M \quad \text{--- (2)}$$

and real valued basis functions $\phi_1(t), \phi_2(t) \dots \phi_N(t)$ forms orthonormal set

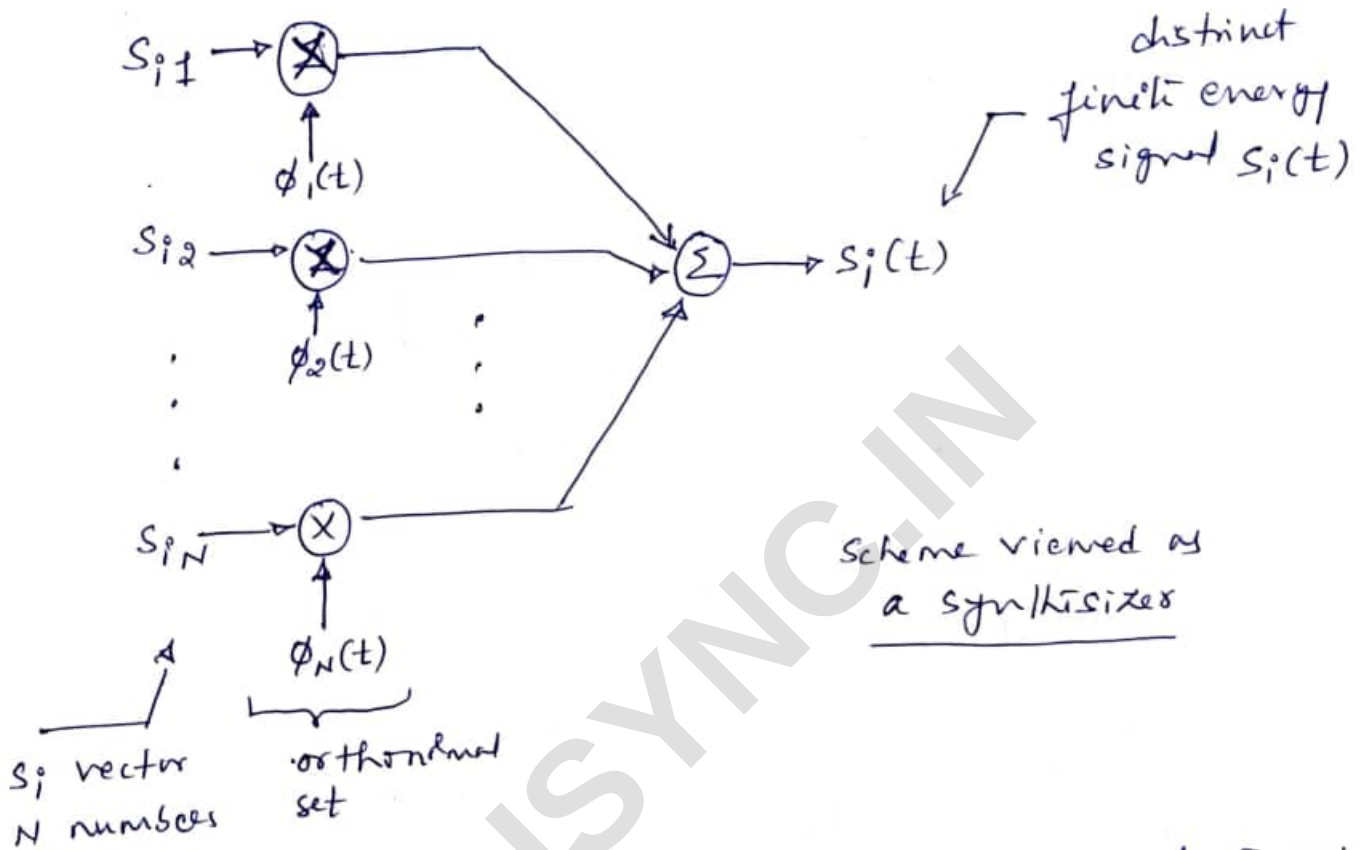
$$\int_0^T \phi_i(t) \phi_j(t) dt = \delta_{ij} = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases} \quad \text{--- (3)}$$

Kronecker delta $\uparrow$

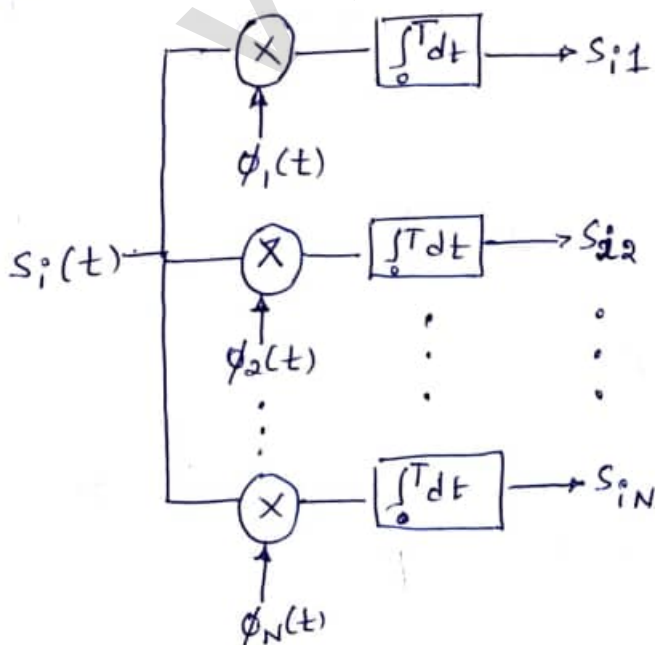
for prescribed i , set of coefficients $\{s_{ij}\}_{j=1}^N$ may be viewed as

N -dimensional signal vector. denoted by S_i .

Given N elements of vector S_i operating as input,
 signal $s_i(t)$ generated from eqn. ①



Also, given signal $s_i(t)$, $i = 1, 2, \dots, M$, operating as input,
 we can calculate coefficients $s_{i1}, s_{i2}, \dots, s_{iN}$ from eqn ②



bank of N -product
 integrators or
 correlators with
 common input viewed
 as analyzer

Relation between length of vector and signal energy.

Let us consider an arbitrary vector S_i , in N -dimensional Euclidian space, squared length of this vector is given by

$$\|S_i\|^2 = (S_i, S_i) = S_i^T S_i \quad \text{--- (4)}$$

$$= [S_{i1}, S_{i2}, \dots, S_{iN}] \begin{bmatrix} S_{i1} \\ S_{i2} \\ \vdots \\ S_{iN} \end{bmatrix} \quad \text{--- (5)}$$

$$= S_{i1}^2 + S_{i2}^2 + \dots + S_{iN}^2 \quad \text{--- (6)}$$

$$\therefore \|S_i\|^2 = \sum_{j=1}^N S_{ij}^2 \quad \text{--- (7)}$$

$i = 1, 2, \dots, M.$

where S_{ij} are co-ordinates of S_i

Let S_i and S_j are two vectors in set, then the cosine of angle between two vectors S_i and S_j is

$$\cos \theta_{ij} = \frac{\text{inner product of } S_i \text{ and } S_j}{(\text{Length of } S_i)(\text{Length of } S_j)} = \frac{S_i^T \cdot S_j}{\|S_i\| \|S_j\|} \quad \text{--- (8)}$$

Now energy E_i associated with signal $S_i(t)$ is

$$E_i = \int_0^T S_i^2(t) dt$$
$$= \int_0^T \left(\sum_{j=1}^N S_{ij} \phi_j(t) \right) \left(\sum_{k=1}^N S_{ik} \phi_k(t) \right) dt \quad \text{--- (9)}$$

Reordering eqn(8),

$$E_i = \sum_{j=1}^N \sum_{k=1}^N s_{ij} s_{ik} \underbrace{\int_0^T \phi_j(t) \phi_k(t) dt}_{\text{using orthogonal property}} \quad \text{--- (9)}$$

using orthogonal property $\int_0^T \phi_j(t) \phi_k(t) dt = \begin{cases} 1 & \text{if } j=k \\ 0 & \text{if } j \neq k \end{cases}$

eqn (9), rewritten as

$$E_i = \sum_{j=1}^N s_{ij}^2 = \|s_i\|^2 \quad \text{--- (10)}$$

thus energy of the signal $s_i(t)$ is equal to square of the length of corresponding vector s_i

Now, Euclidean distance between s_i and s_j vectors representing signal $s_i(t)$ and $s_j(t)$ is

$$d_{ij} = \|s_i - s_j\| = \left[\sum_{k=1}^N (s_{ik} - s_{jk})^2 \right]^{1/2} \quad \text{--- (11)}$$

also $d_{ij} = \left[\int_0^T (s_i(t) - s_j(t))^2 dt \right]^{1/2} \quad \text{--- (12)}$

Gram Schmidt Orthogonalisation procedure.

Real valued energy signal $s_1(t), s_2(t) \dots s_m(t)$
each of duration T seconds represented by

$$s_i(t) = \sum_{j=1}^N s_{ij} \phi_j(t), \quad \begin{cases} 0 \leq t \leq T \\ i=1, 2, \dots, M \end{cases} \quad (1)$$

where co-efficient of expansion defined by

$$s_{ij} = \int_0^T s_i(t) \phi_j(t) dt, \quad \begin{cases} i=1, 2, \dots, M \\ j=1, 2, \dots, N \end{cases} \quad (2)$$

and real valued basis functions $\phi_1(t), \phi_2(t) \dots \phi_N(t)$ forms orthonormal set, given by

$$\int_0^T \phi_i(t) \phi_j(t) dt = \begin{cases} 1, & i=j \\ 0, & i \neq j \end{cases} \quad (3)$$

$\therefore$ eqn (1) can be expanded as

$$\left. \begin{aligned} s_1(t) &= s_{11} \phi_1(t) + s_{12} \phi_2(t) + \dots + s_{1N} \phi_N(t) \\ s_2(t) &= s_{21} \phi_1(t) + s_{22} \phi_2(t) + \dots + s_{2N} \phi_N(t) \\ s_3(t) &= s_{31} \phi_1(t) + s_{32} \phi_2(t) + \dots + s_{3N} \phi_N(t) \\ &\vdots \\ s_m(t) &= s_{m1} \phi_1(t) + s_{m2} \phi_2(t) + \dots + s_{mN} \phi_N(t) \end{aligned} \right\} \begin{array}{l} 4.1 \\ 4.2 \\ 4.3 \\ \textcircled{4} \\ 4.N \end{array}$$

to find orthonormal set of basis functions.

Step 1: put $s_{ij} = 0$, except s_{11} in eqn 4.1

$$s_1(t) = s_{11} \phi_1(t) \quad \text{--- (5)}$$

$$\therefore \boxed{\phi_1(t) = \frac{s_1(t)}{s_{11}}} \quad \text{--- (6)}$$

where $s_{11} = \sqrt{E_1}$ energy of signal $s_1(t)$

to find s_{11} , square eqn 4.1 both sides and integrate over T seconds.

$$\int_0^T s_1^2(t) dt = \int_0^T s_{11}^2 \phi_1^2(t) dt \quad \text{--- (7)}$$

$\underbrace{\quad}_{=1} \text{ as } i=j$
having unit magnitude

$$\therefore \boxed{s_{11} = \sqrt{\int_0^T s_1^2(t) dt}} \quad \text{--- (8)}$$

Step 2: consider eqn 4.2.

$$s_2(t) = s_{21} \phi_1(t) + s_{22} \phi_2(t) \quad \text{--- 4.2}$$

multiply both sides by $\phi_1(t)$ and integrate over T seconds

$$\therefore \int_0^T s_{21} \underbrace{\phi_1^2(t)}_{=1, i=j} dt + \int_0^T s_{22} \underbrace{\phi_1(t) \phi_2(t)}_{=0, i \neq j} dt = \int_0^T s_2(t) \phi_1(t) dt$$

$$\therefore \boxed{s_{21} = \int_0^T s_2(t) \phi_1(t) dt} \quad \text{--- (9)}$$

Now let $g_2(t) = s_2(t) - s_{21}\phi_1(t)$ — (10)

where $g_2(t) = s_{22}\phi_2(t)$ — (11)

from eqn (11), squaring both sides & integrating over T seconds.

$$\int_0^T \underbrace{s_{22}^2}_{=1, \text{ for } i=j} \phi_2^2(t) dt = \int_0^T g_2^2(t) dt$$

$$\therefore \boxed{s_{22} = \sqrt{\int_0^T g_2^2(t) dt}} \quad (12)$$

from eqn (11),

$$\therefore \boxed{\phi_2(t) = \frac{g_2(t)}{\sqrt{\int_0^T g_2^2(t) dt}}} \quad (13)$$

Continuing this procedure, we may, in general define.

$$g_i(t) = s_i(t) - \sum_{j=1}^{i-1} s_{ij}\phi_j(t), \quad j=1, 2, \dots, i-1 \quad (14)$$

$$s_{ij} = \int_0^T s_i(t) \phi_j(t) dt, \quad j=1, 2, \dots, i-1 \quad (15)$$

$$\phi_i(t) = \frac{g_i(t)}{\sqrt{\int_0^T g_i^2(t) dt}}, \quad j=1, 2, \dots, N \quad (16)$$

Summary.

① $\phi_i(t) = \frac{s_i(t)}{\sqrt{E_i}}$ where $\sqrt{E_i} = \int_0^T s_i^2(t) dt = S_{ii}$
for $i=1$

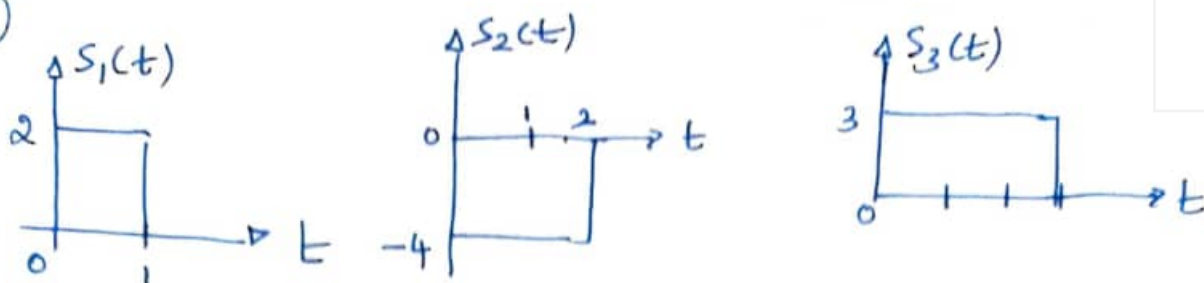
② Consider intermediate function $g_i(t)$
$$g_i(t) = s_i(t) - \sum_{j=1}^{i-1} s_{ij} \phi_j(t) \quad j=1, 2, \dots, i-1$$

Where
$$s_{ij} = \int_0^T s_i(t) \phi_j(t) dt, \quad j=1, 2, \dots, N$$

$$\phi_i(t) = \frac{g_i(t)}{\sqrt{\int_0^T g_i^2(t) dt}}$$

③ for $i=2, 3, \dots, M-1$,
$$\sqrt{E_i} = \int_0^T g_i^2(t) dt \Rightarrow S_{22}, S_{33}, S_{44}, \dots$$

①



using Gram Schmidt orthogonalisation, find set of orthonormal basis functions.

finite energy signal for T duration $S_i(t)$ is given by

$$S_i(t) = \sum_{j=1}^N S_{ij} \phi_j(t), \quad \begin{cases} 0 \leq t \leq T \\ i = 1, 2, \dots, M \end{cases} \quad (1)$$

where

$$S_{ij} = \int_0^T S_i(t) \phi_j(t) dt, \quad j = 1, 2, \dots, N$$

$$\text{and } \int_0^T \phi_i(t) \phi_j(t) dt = \begin{cases} 1, & \text{for } i=j \\ 0, & \text{for } i \neq j \end{cases}$$

$\therefore$ eqn (1) can be expanded as

$$S_1(t) = S_{11} \phi_1(t)$$

$$S_2(t) = S_{21} \phi_1(t) + S_{22} \phi_2(t)$$

$$S_3(t) = S_{31} \phi_1(t) + S_{32} \phi_2(t) + S_{33} \phi_3(t)$$

Step 1: first orthonormal basis function for $i=1$

$$\therefore \phi_1(t) = \frac{S_1(t)}{S_{11}} = \frac{S_1(t)}{\sqrt{E_1}}$$

$$\begin{aligned} \text{but } S_{11} &= \sqrt{\int_0^T S_1^2(t) dt} = \sqrt{E_1} \\ &= \left(\int_0^1 2^2 dt \right)^{1/2} = 2 \end{aligned}$$

$$\therefore \phi_1(t) = \frac{1}{2} S_1(t) = \begin{cases} 1, & 0 \leq t \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

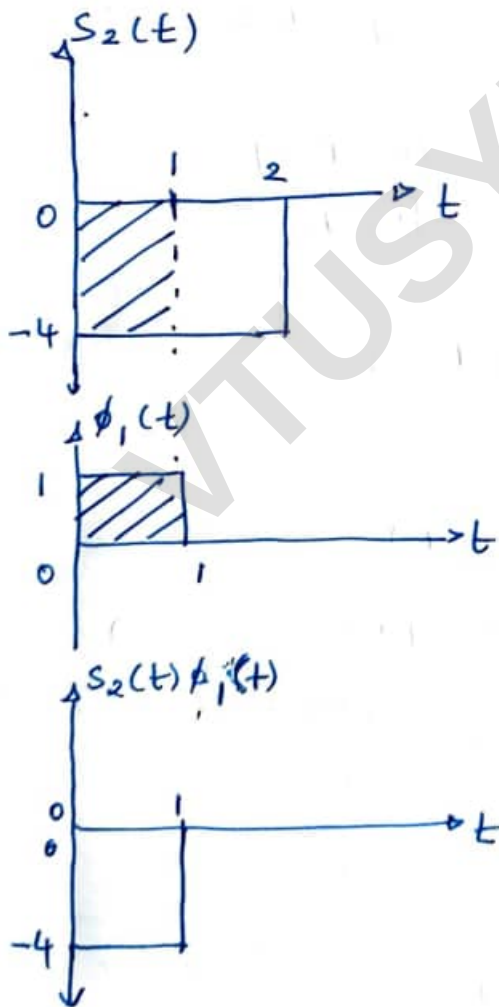
Step 2: for $i = 2$,

$$S_2(t) = S_{21} \phi_1(t) + S_{22} \phi_2(t)$$

$$\text{Let } g_2(t) = S_{22} \phi_2(t) = S_2(t) - S_{21} \phi_1(t)$$

$$\text{but } S_{21} = \int_0^T S_2(t) \phi_1(t) dt$$

find product of the signal and integration range

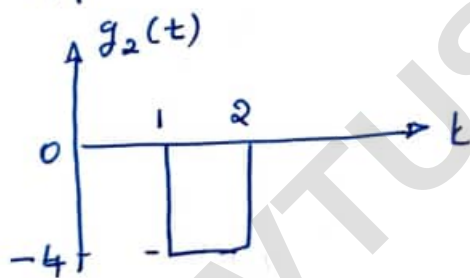
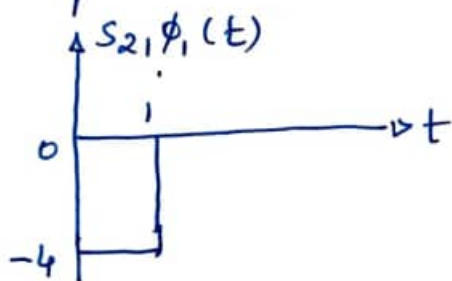
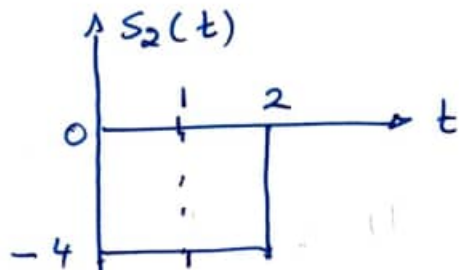


$$\therefore S_{21} = \int_0^1 (-4) dt = \underline{\underline{-4}}$$

Now $g_2(t) = s_{22} \phi_2(t)$

$$\therefore s_{22} = \left(\int_0^T g_2^2(t) dt \right)^{1/2}$$

but $g_2(t) = s_2(t) - s_{21} \phi_1(t)$



$$\therefore s_{22} = \left(\int_1^2 (-4)^2 dt \right)^{1/2} = \sqrt{16} = \underline{4}$$

$$\therefore \phi_2(t) = \frac{g_2(t)}{s_{22}} = \frac{1}{4} s_2(t)$$

$$\phi_2(t) = \frac{g_2(t)}{4} = \begin{cases} -1, & 1 \leq t \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

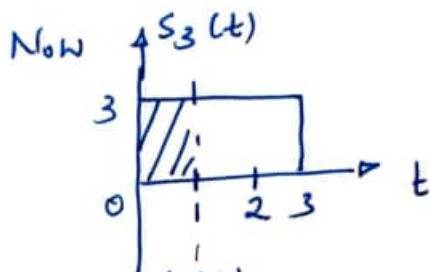
step 3: for $j=3$

$$s_3(t) = s_{31} \phi_1(t) + s_{32} \phi_2(t) + s_{33} \phi_3(t)$$

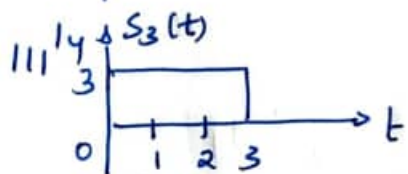
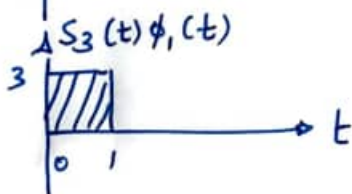
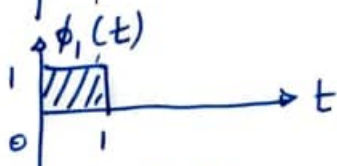
Let $g_3(t) = s_{33} \phi_3(t) = s_3(t) - s_{31} \phi_1(t) - s_{32} \phi_2(t)$

but $s_{31} = \int_0^T s_3(t) \phi_1(t) dt$

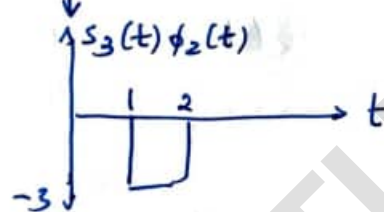
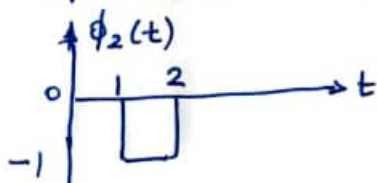
$$s_{32} = \int_0^T s_{32} \phi_2(t) dt$$



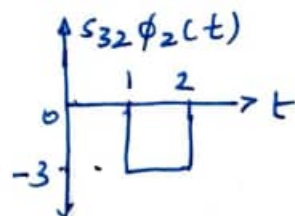
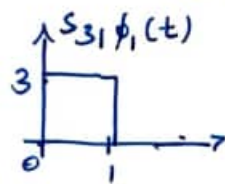
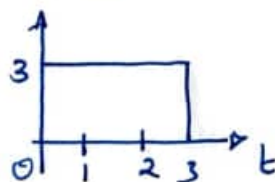
$$S_{31} = \int_0^1 3 \cdot dt = \underline{\underline{3}}$$



$$S_{32} = \int_1^2 -3 \cdot dt = \underline{\underline{-3}}$$



Now $g_3(t) = S_3(t) - S_{31}\phi_1(t) - S_{32}\phi_2(t)$



We can find from these signals

$$\left. \begin{aligned} g_3(t) \Big|_{t=0 \text{ to } 1} &= 3 - 3 - 0 = 0 \\ g_3(t) \Big|_{t=1 \text{ to } 2} &= 3 - 0 - 3 = 0 \\ g_3(t) \Big|_{t=2 \text{ to } 3} &= 3 - 0 - 0 = 3 \end{aligned} \right\} \therefore g_3(t) = \begin{cases} 0, & 0 \leq t \leq 1 \\ 0, & 1 \leq t \leq 2 \\ 3, & 2 \leq t \leq 3 \end{cases}$$

$$\therefore S_{33} = \sqrt{\int_0^T g_3^2(t) dt} = \sqrt{\int_2^3 3^2 \cdot dt} = \underline{\underline{3}}$$

$$\phi_3(t) = \frac{g_3(t)}{S_{33}} = \frac{g_3(t)}{\underline{\underline{3}}} = \begin{cases} 1, & 2 \leq t \leq 3 \\ 0, & 0 \leq t \leq 2, \\ & t > 3 \end{cases}$$

$\therefore$ the orthonormal basis functions found to be

$$\phi_1(t) = \frac{S_1(t)}{2}, \quad \phi_2(t) = \frac{g_2(t)}{4}, \quad \phi_3(t) = \frac{g_3(t)}{3}$$

$$\therefore S_1(t) = S_{11} \phi_1(t) = 2 \phi_1(t)$$

$$S_2(t) = -4 \phi_1(t) + 4 \phi_2(t)$$

$$S_3(t) = 3 \phi_1(t) - 3 \phi_2(t) + 3 \phi_3(t)$$

Remember:

Step 1. for $i=1$, $\phi_1(t) = \frac{S_1(t)}{S_{11}}$, $S_{11} = \sqrt{E_1} = \left(\int_0^T S_1^2(t) dt \right)^{1/2}$

Step 2. for $i=2$,

$$g_i(t) = S_i(t) - \sum_{j=1}^{i-1} S_{ij} \phi_j(t)$$

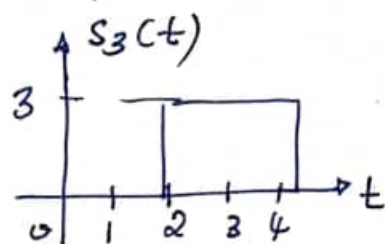
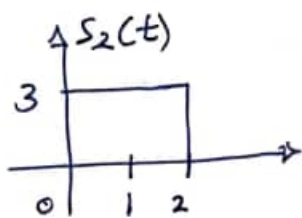
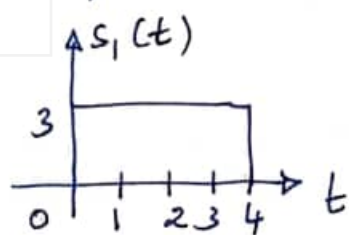
where $S_{ij} = \int_0^T S_i(t) \phi_j(t) dt$

Step 3: $\phi_i(t) = \frac{g_i(t)}{\sqrt{\int_0^T g_i^2(t) dt}}$

where $S_{22}, S_{33}, S_{44} \dots = \sqrt{E_i}$

$$= \sqrt{\int_0^T g_i^2(t) dt}$$

② given $S_1(t)$, $S_2(t)$, $S_3(t)$ obtain orthogonalisation function using Gram Schmidt procedure.



energy signal $S_1(t)$, $S_2(t)$, ..., $S_M(t)$ for T seconds represented by

$$S_i(t) = \sum_{j=1}^N S_{ij} \phi_j(t), \quad \begin{cases} 0 \leq t \leq T \\ i=0, 1, 2, \dots, M \end{cases}$$

step 1: to find S_{11} & $\phi_1(t)$, for $i=1$

$$S_1(t) = S_{11} \phi_1(t)$$

$$\therefore \phi_1(t) = \frac{S_1(t)}{S_{11}}, \quad S_{11} = \left[\int_0^T S_1^2(t) dt \right]^{1/2}$$

$$\therefore S_{11} = \left[\int_0^4 3^2 dt \right]^{1/2} = [9 \cdot 4]^{1/2} = \underline{\underline{6 = S_{11}}}$$

$$\boxed{\phi_1(t) = \frac{S_1(t)}{6}} = \begin{cases} 1/2, & 0 \leq t \leq 4 \\ 0, & \text{otherwise.} \end{cases}$$

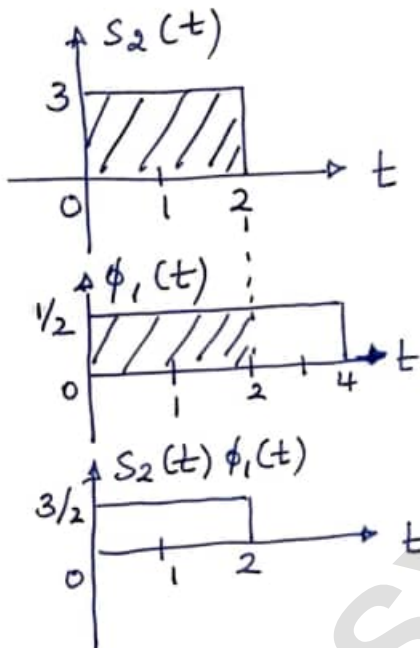
step 2 for $i=2$, let us find $\phi_2(t)$.

$$S_2(t) = S_{21} \phi_1(t) + S_{22} \phi_2(t)$$

$$\text{let } g_2(t) = s_{22} \phi_2(t)$$

$$\therefore s_{22}^2 = \int_0^T g_2^2(t) dt$$

$$\text{but } g_2(t) = s_2(t) - s_{21} \phi_1(t)$$



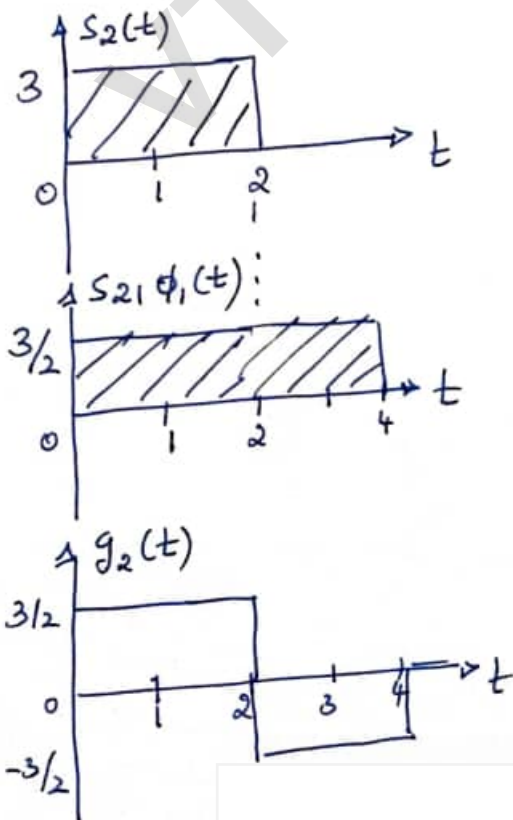
$$\text{but } s_{21} = \int_0^T s_2(t) \phi_1(t) dt$$

$$\therefore s_{21} = \int_0^2 3/2 \cdot dt$$

$$= 3/2 \times 2 = 3$$

$$\boxed{s_{21} = 3}$$

$$\text{Now, } g_2(t) = s_2(t) - s_{21} \phi_1(t)$$



$$g_2(t) \Big|_{t=0 \text{ to } 2} = 3 - 3/2 = 3/2$$

$$g_2(t) \Big|_{t=2 \text{ to } 4} = 0 - 3/2 = -3/2$$

$$\therefore g_2(t) = \begin{cases} 3/2, & 0 \leq t < 2 \\ -3/2, & 2 \leq t < 4 \\ 0, & t > 4 \end{cases}$$

Now,

$$\begin{aligned} S_{22} &= \int_0^T g_2^2(t) dt \\ &= \int_0^2 \left(\frac{3}{2}\right)^2 dt + \int_2^4 \left(-\frac{3}{2}\right)^2 dt \\ &= \frac{9}{4} \times 2 + \frac{9}{4} \times 2 = \underline{9} \end{aligned}$$

$$\boxed{S_{22} = 3}$$

$$\therefore g_2(t) = S_{22} \phi_2(t)$$

$$\phi_2(t) = \frac{g_2(t)}{S_{22}} = \frac{g_2(t)}{3} = \begin{cases} \frac{1}{2}, & 0 \leq t \leq 2 \\ -\frac{1}{2}, & 2 \leq t \leq 4 \\ 0, & t > 4 \end{cases}$$

Step 3: for $i=3$, let us find $\phi_3(t)$

$$s_3(t) = S_{31} \phi_1(t) + S_{32} \phi_2(t) + S_{33} \phi_3(t)$$

$$\text{Here, } S_{31} = \int_0^T s_3(t) \phi_1(t) dt$$

$$S_{32} = \int_0^T s_3(t) \phi_2(t) dt$$

$$g_3(t) = S_{33} \phi_3(t) = s_3(t) - S_{31} \phi_1(t) - S_{32} \phi_2(t)$$

$$\therefore S_{33} = \sqrt{\int_0^T g_3^2(t) dt}, \quad \phi_3(t) = \frac{g_3(t)}{S_{33}}$$

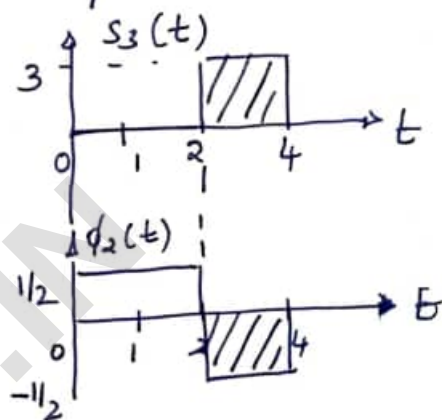
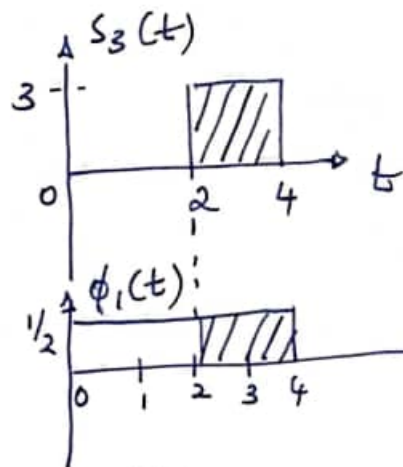
Now,

$$S_{31} = \int_2^4 3 \cdot \frac{1}{2} \cdot dt = \frac{3}{2} \times 2$$

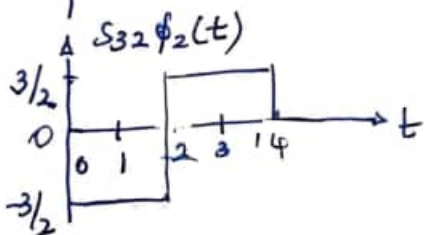
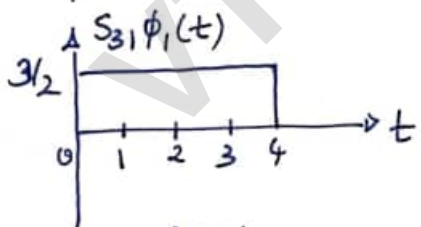
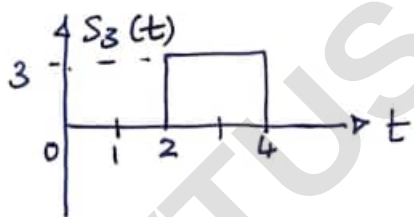
$$\boxed{S_{31} = 3}$$

$$S_{32} = \int_2^4 3 \cdot \left(-\frac{1}{2}\right) dt = -\frac{3}{2} \times 2$$

$$\boxed{S_{32} = -3}$$



Now, $g_3(t) = S_3(t) - S_{31}\phi_1(t) - S_{32}\phi_2(t)$



$$g_3(t) \Big|_{t=0 \text{ to } 2} = 0 - \frac{3}{2} + \frac{3}{2} = 0$$

$$g_3(t) \Big|_{t=2 \text{ to } 4} = 3 - \frac{3}{2} - \frac{3}{2} = 0$$

$$\therefore \boxed{S_{33} = 0}$$

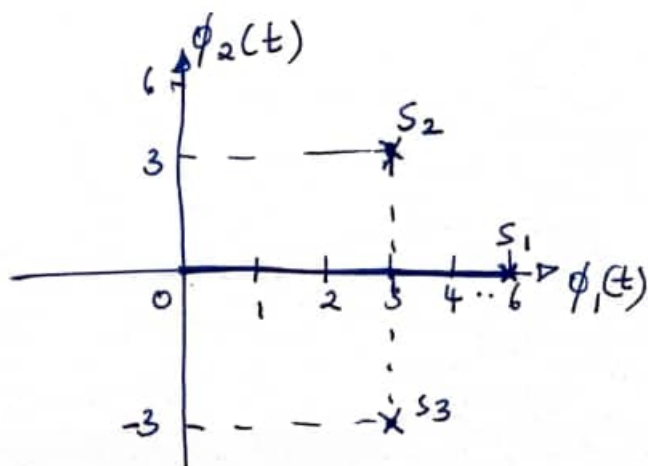
$$\phi_3(t) = \frac{g_3(t)}{S_{33}}$$

not exist

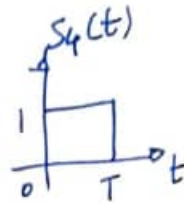
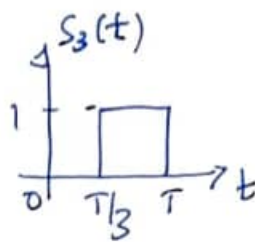
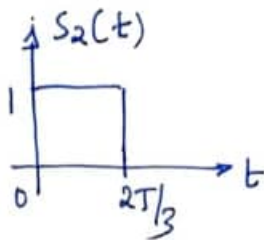
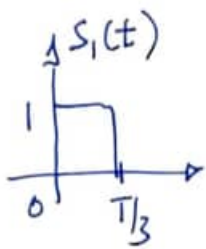
$$\therefore S_1(t) = 6\phi_1$$

$$S_2(t) = 3\phi_1 + 3\phi_2$$

$$S_3(t) = 3\phi_1 - 3\phi_2$$



③ using Gram Schmidt orthogonalisation determine orthonormal basis functions for the following signals.



Solution:

Step 1: $S_i(t) = \sum_{j=1}^N s_{ij} \phi_j(t), \quad \begin{cases} 0 \leq t \leq T \\ \text{for } i=1, 2, \dots, M \end{cases} \quad \text{①}$

Where $s_{ij} = \int_0^T S_i(t) \phi_j(t) dt$ $\begin{cases} i=1, 2, \dots, M \\ j=1, 2, \dots, N \end{cases}$

and $\int_0^T \phi_j(t) \phi_i(t) dt = \begin{cases} 1, & i=j \\ 0, & i \neq j \end{cases}$

for $i=1$, eqn ①

$$S_1(t) = s_{11} \phi_1(t)$$

$$\therefore \phi_1(t) = \frac{S_1(t)}{s_{11}}$$

$$\begin{aligned} \therefore \phi_1(t) &= \frac{S_1(t)}{\sqrt{T/3}} \\ &= \frac{\sqrt{3} S_1(t)}{\sqrt{T}} \end{aligned}$$

$$\boxed{\phi_1(t) = \begin{cases} \sqrt{3/T}, & 0 \leq t \leq T/3 \\ 0, & \text{otherwise} \end{cases}}$$

Where $s_{11} = \sqrt{\int_0^T S_1^2(t) dt}$

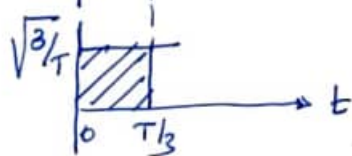
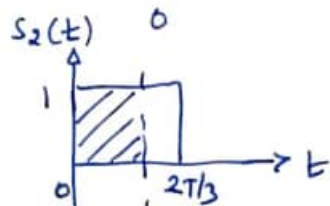
$$s_{11} = \sqrt{\int_0^{T/3} 1^2 dt}$$

$$\boxed{s_{11} = \sqrt{T/3}}$$

Step 2: for $i=2$, eqn ① is

$$s_2(t) = s_{21}\phi_1(t) + s_{22}\phi_2(t)$$

$$s_{21} = \int_0^T s_2(t)\phi_1(t) dt$$



$$s_{21} = \int_0^{T/3} 1 \cdot \sqrt{3}/T dt$$

$$= \sqrt{3} \cdot T/3 = \sqrt{T/3}$$

$$\boxed{s_{21} = \sqrt{T/3}}$$

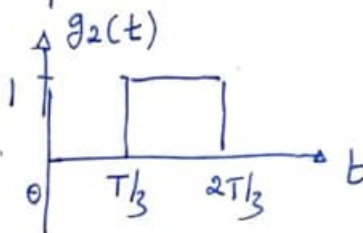
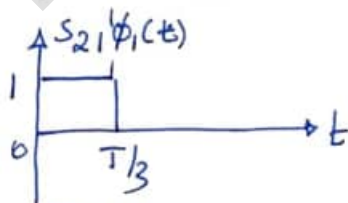
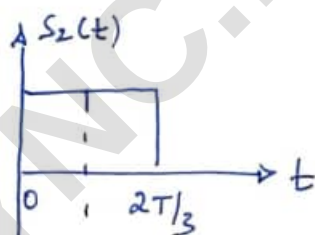
$$g_2(t) = s_{22}\phi_2(t)$$

$$s_{22} = \left(\int_0^T g_2^2(t) dt \right)^{1/2}$$

$$g_2(t) = s_2(t) - s_{21}\phi_1(t)$$

$$\phi_2(t) = \frac{g_2(t)}{s_{22}}$$

$$g_2(t) = s_2(t) - s_{21}\phi_1(t)$$



$$s_{21}\phi_1(t) = \sqrt{T/3}\phi_1(t)$$

$$= \begin{cases} 1, & T/3 \leq t \leq 2T/3 \\ 0, & \text{otherwise} \\ 0 \leq t \leq T/3 \end{cases}$$

$$\therefore s_{22} = \left(\int_{T/3}^{2T/3} g_2^2(t) dt \right)^{1/2}$$

$$\boxed{s_{22} = \sqrt{T/3}}$$

$$\therefore \phi_2(t) = \frac{g_2(t)}{\sqrt{T/3}} = \sqrt{\frac{3}{T}} g_2(t), \begin{cases} \sqrt{3}/T, & T/3 \leq t \leq 2T/3 \\ 0, & 0 \leq t \leq T/3 \end{cases}$$

$$\begin{cases} \sqrt{3}/T, & T/3 \leq t \leq 2T/3 \\ 0, & 0 \leq t \leq T/3 \end{cases}$$

Step 3: for $i=3$

$$s_3(t) = s_{31}\phi_1(t) + s_{32}\phi_2(t) + s_{33}\phi_3(t)$$

here.

$$s_{31} = \int_0^T s_3(t)\phi_1(t) dt$$

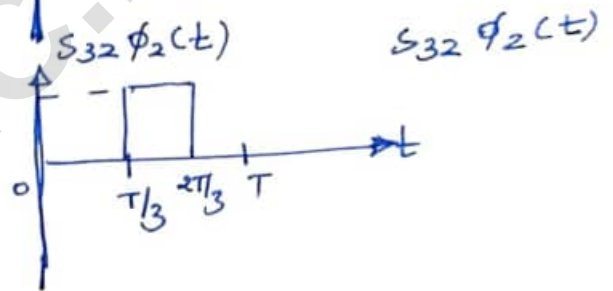
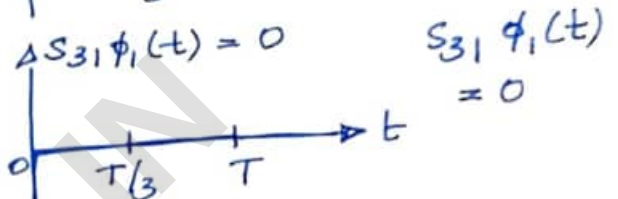
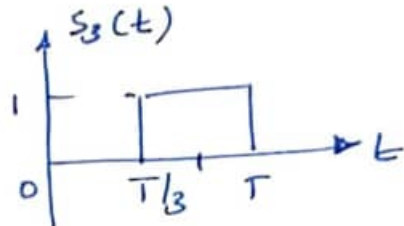
$$s_{32} = \int_0^T s_3(t)\phi_2(t) dt$$

$$g_3(t) = s_{33}\phi_3(t)$$

$$s_{33} = \sqrt{\int_0^T g_3^2(t) dt}$$

$$\phi_3(t) = \frac{g_3(t)}{s_{33}}$$

$$g_3(t) = s_3(t) - s_{31}\phi_1(t) - s_{32}\phi_2(t)$$



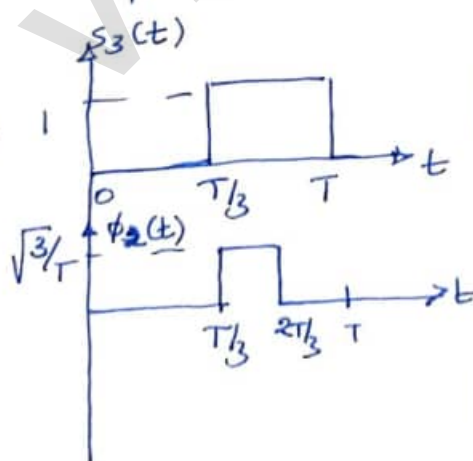
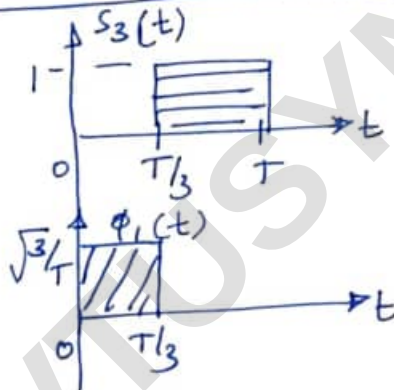
$$g_3(t) \Big|_{0 \leq t < T/3} = 0 - 0 - 0 = 0$$

$$g_3(t) \Big|_{T/3 \leq t < 2T/3} = 1 - 0 - 1 = 0$$

$$g_3(t) \Big|_{2T/3 \leq t \leq T} = 1 - 0 - 0 = 1$$

$$g_3(t) = \begin{cases} 1 & 2T/3 \leq t \leq T \\ 0 & 0 \leq t \leq 2T/3, t > T \end{cases}$$

$$s_{31} = 0$$



$$s_{32} = \int_{T/3}^{2T/3} 1 \cdot \frac{\sqrt{3}}{T} dt$$

$$= \frac{\sqrt{3}}{T} \cdot \frac{T}{3}$$

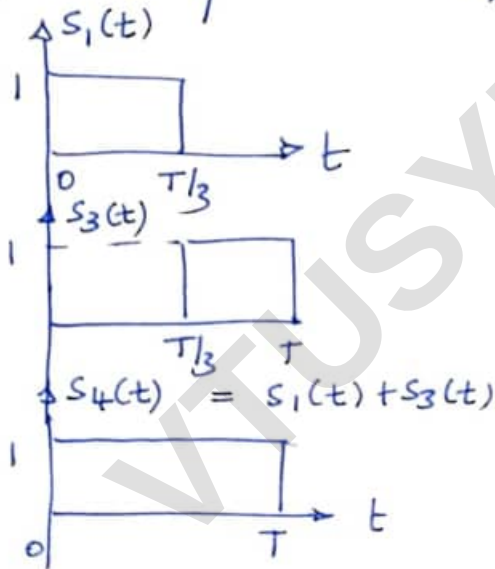
$$s_{32} = \frac{\sqrt{T}}{3}$$

$$S_{33} = \sqrt{\int_{2T/3}^T g_3^2(t) dt} = \sqrt{\left(T - 2T/3\right)} = \sqrt{T/3}$$

$$S_{33} = \sqrt{T/3}$$

$$\phi_3 = \frac{S_3(t)}{S_{33}} = \sqrt{\frac{3}{T}} \cdot S_3(t) \quad \begin{cases} \sqrt{3}/4, & T/3 \leq t \leq T \\ 0, & t > T, 0 \leq t \leq T/3 \end{cases}$$

Step 4: Now from waveforms given,

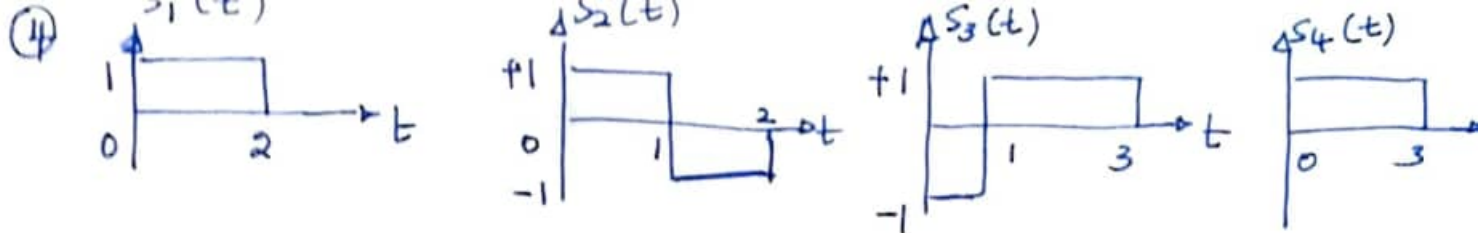


$$\therefore S_1(t) = \sqrt{T/3} \phi_1(t)$$

$$S_2(t) = \sqrt{T/3} \phi_1(t) + \sqrt{T/3} \phi_2(t)$$

$$S_3(t) = \sqrt{T/3} \phi_2(t)$$

$$S_4(t) = \sqrt{T/3} \phi_1(t) + \sqrt{T/3} \phi_2(t)$$



Step 1: for $i=1$, find S_{11} , $\phi_1(t)$

$$S_{11} = \sqrt{\int_0^T S_1^2(t) dt} = \sqrt{\int_0^2 1 \cdot dt} = \sqrt{2}$$

$$\phi_1(t) = \frac{S_1(t)}{S_{11}}, \quad \begin{cases} 1/\sqrt{2}, & 0 \leq t \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

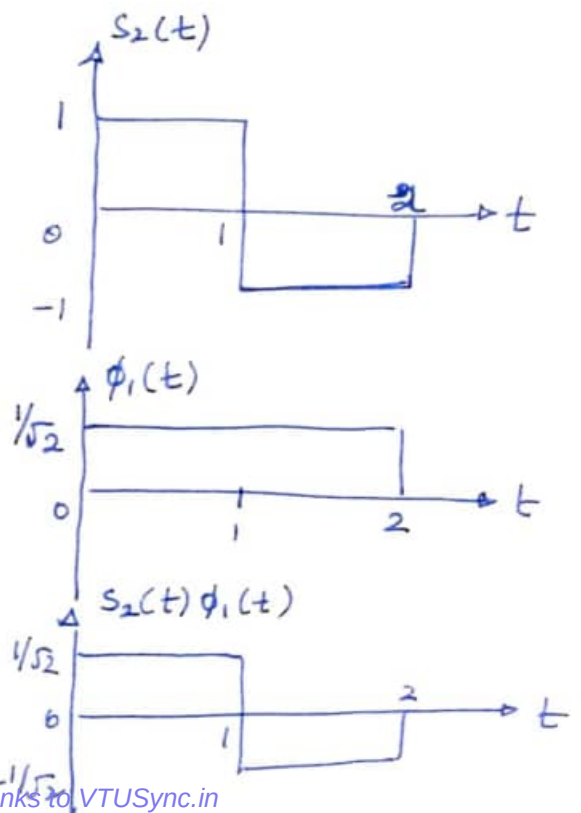
Step 2: for $i=2$, find S_{21} , S_{22} , $\phi_2(t)$ | $\phi_1(t)$ is known

$$S_{21} = \int_0^T S_2(t) \phi_1(t) dt \quad g_2(t) = S_2(t) - S_{21} \phi_1(t)$$

$$S_{22} = \sqrt{\int_0^T g_2^2(t) \cdot dt}, \quad \phi_2(t) = \frac{g_2(t)}{S_{22}}$$

$$\begin{aligned} S_{21} &= \int_0^T S_2(t) \phi_1(t) \cdot dt \\ &= \int_0^1 1/\sqrt{2} \cdot dt + \int_1^2 (-1/\sqrt{2}) dt \end{aligned}$$

$$S_{21} = 0$$



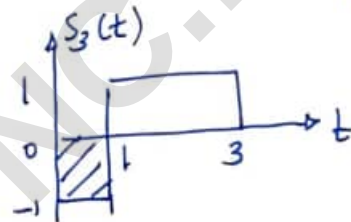
$$\therefore g_2(t) = s_2(t), \quad \begin{cases} 1, & 0 \leq t \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

$$\therefore \left[S_{22} = \sqrt{\int_0^1 g_2^2(t) dt} = \sqrt{\int_0^1 1 \cdot dt} = 1 \right]$$

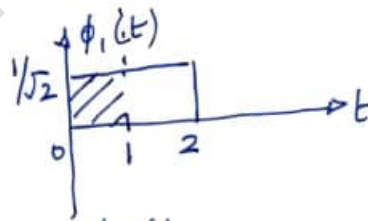
$$\left[\phi_2(t) = \frac{g_2(t)}{S_{22}} = g_2(t), \quad \begin{cases} 1, & 0 \leq t \leq 1 \\ 0, & \text{otherwise} \end{cases} \right]$$

Step 3: for $i=3$, $s_{31}, s_{32}, s_{33}, \phi_3(t)$ | $\phi_1(t), \phi_2(t)$ are known.

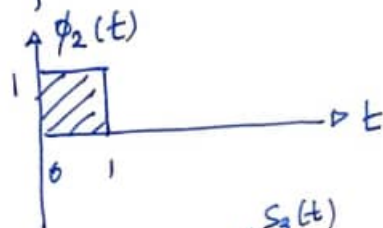
$$s_{31} = \int_0^T s_3(t) \phi_1(t) dt$$



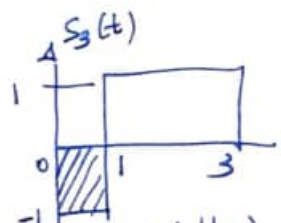
$$s_{32} = \int_0^T s_3(t) \phi_2(t) dt$$



$$s_{31} = \int_0^1 -1 \cdot 1/\sqrt{2} dt = \underline{\underline{-1/\sqrt{2}}}$$



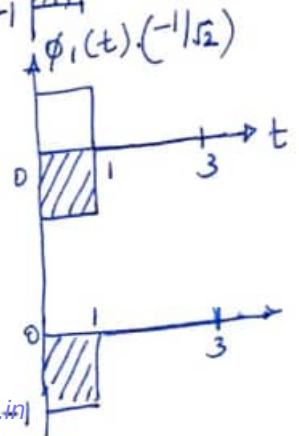
$$s_{32} = \int_0^1 1 \cdot (-1/\sqrt{2}) dt = \underline{\underline{-1}}$$



$$g_3(t) = s_3(t) - s_{31}\phi_1(t) - s_{32}\phi_2(t)$$

$$g_3(t) \Big|_{t=0 \text{ to } 1} = -1 + 1/\sqrt{2} + 1 = 1/\sqrt{2} \quad 0 \leq t \leq 1$$

$$g_3(t) \Big|_{t=1 \text{ to } 3} = 1 - 0 - 0 = 1, \quad 1 \leq t \leq 3$$



$$g_3(t) = \begin{cases} 1/\sqrt{2}, & 0 \leq t \leq 1 \\ 1, & 1 \leq t \leq 3 \end{cases}$$

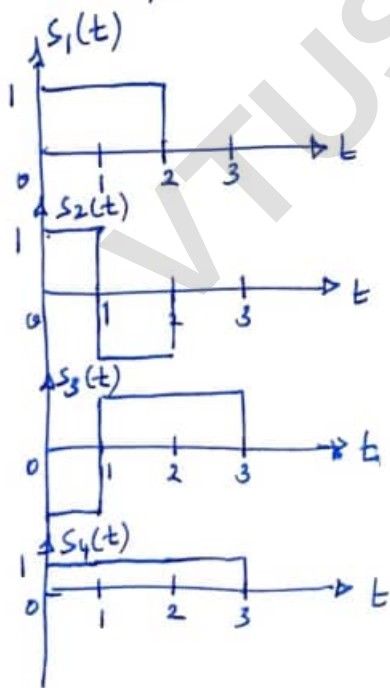
$$S_{33} = \sqrt{\int_0^T g_3^2(t) dt} = \sqrt{\int_0^1 (1/\sqrt{2})^2 dt + \int_1^3 1^2 dt}$$

$$= \sqrt{1/2 + 2} = \underline{\underline{\sqrt{5/2}}}$$

$$\phi_3(t) = \frac{g_3(t)}{S_{33}} = \sqrt{2/5} g_3(t) = \begin{cases} 1/\sqrt{5}, & 0 \leq t \leq 1 \\ \sqrt{2/5}, & 1 \leq t \leq 3 \end{cases}$$

Step 4: by observing the waveforms,

$$S_4(t) = S_1(t) + S_2(t) + S_3(t)$$



$$\therefore S_1(t) = \sqrt{2} \phi_1(t)$$

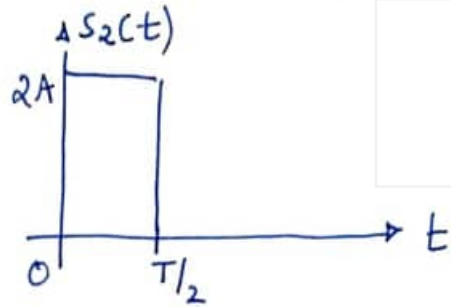
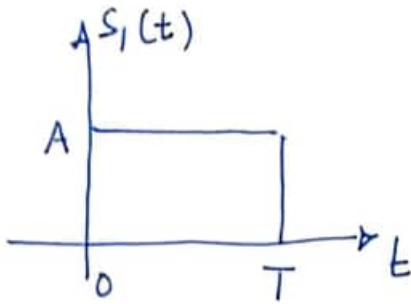
$$S_2(t) = 1 \cdot \phi_2(t)$$

$$S_3(t) = -1/\sqrt{2} \phi_1(t) - \phi_2(t) + \sqrt{5/2} \phi_3(t)$$

$$S_4(t) = \sqrt{2} \phi_1(t) + \phi_2(t) - 1/\sqrt{2} \phi_1(t) - \phi_2(t) + \sqrt{5/2} \phi_3(t)$$

$$S_4(t) = \sqrt{2} \phi_1(t) - 1/\sqrt{2} \phi_1(t) + \sqrt{5/2} \phi_3(t)$$

- ⑤ given $S_1(t)$ and $S_2(t)$, using G.S. procedure find $\phi_1(t)$ and $\phi_2(t)$ and sketch $\phi_1(t)$ & $\phi_2(t)$.



Solution:

$$S_{11} = \sqrt{\int_0^T S_1^2(t) dt} = \sqrt{\int_0^T A^2 dt} = A\sqrt{T}$$

$$\phi_1(t) = \frac{S_1(t)}{S_{11}} = \frac{A}{A\sqrt{T}} = \frac{1}{\sqrt{T}}, \quad \begin{cases} 1/\sqrt{T}, & 0 \leq t \leq T \\ 0, & \text{otherwise} \end{cases}$$

$$S_{21} = \int_0^T S_2(t) \phi_1(t) dt$$

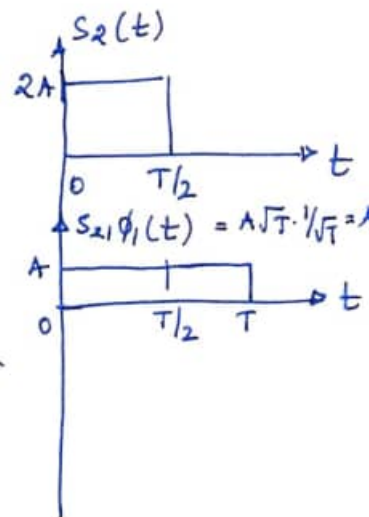
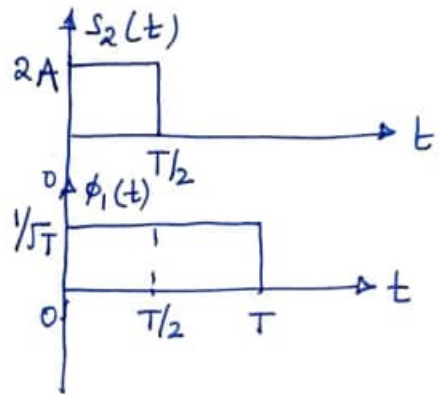
$$= \int_0^{T/2} 2A \cdot \frac{1}{\sqrt{T}} dt = \frac{2A}{\sqrt{T}} \cdot \frac{T}{2}$$

$$S_{21} = \underline{\underline{A\sqrt{T}}}$$

$$S_{22} = \left[\int_0^T (S_2(t) - S_{21}\phi_1(t))^2 dt \right]^{1/2}$$

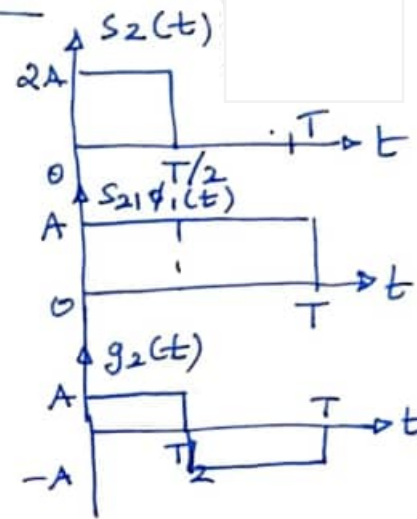
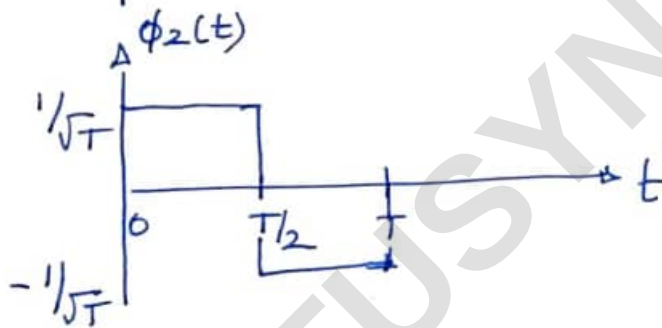
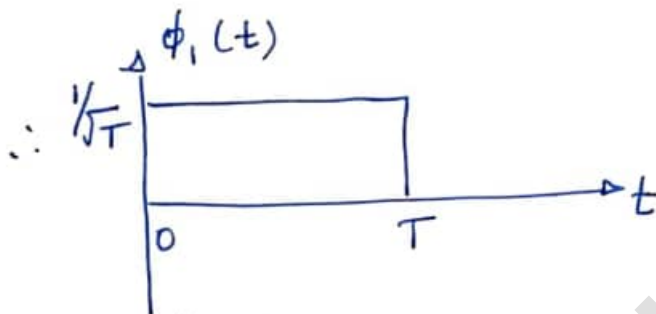
$$= \left[\int_0^{T/2} (2A - A)^2 dt + \int_{T/2}^T (0 - A)^2 dt \right]^{1/2}$$

$$= \left[\int_0^{T/2} A^2 dt + \int_{T/2}^T A^2 dt \right]^{1/2} = \underline{\underline{A\sqrt{T}}}$$

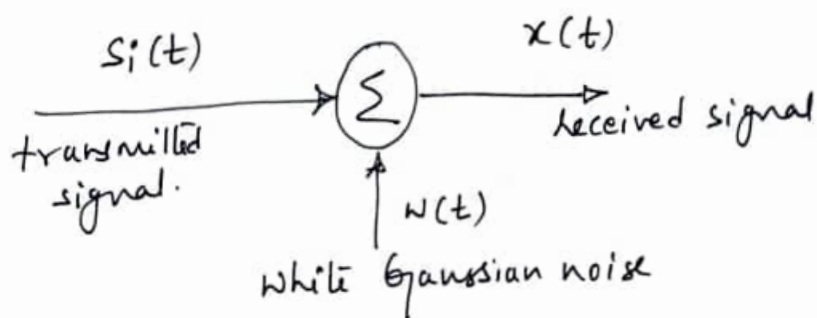


$$\phi_2(t) = \frac{g_2(t)}{s_{22}} = \frac{s_2(t) - s_{21}\phi_1(t)}{s_{22}}$$

$$\phi_2(t) = \begin{cases} 1/\sqrt{T}, & 0 \leq t \leq T/2 \\ -1/\sqrt{T}, & T/2 \leq t \leq T \end{cases}$$



Conversion of continuous AWGN channel into a Vector channel.



set of M -energy signals $s_i(t)$ is linear combination of N -orthonormal basis functions

$$s_i(t) = \sum_{j=1}^N s_{ij} \phi_j(t), \quad \begin{cases} 0 \leq t \leq T \\ i = 1, 2, 3, \dots, M \end{cases} \quad \text{--- (1)}$$

where coefficient of expansion given by

$$s_{ij} = \int_0^T s_i(t) \phi_j(t) dt, \quad \begin{cases} i = 1, 2, \dots, M \\ j = 1, 2, \dots, N \end{cases} \quad \text{--- (2)}$$

the real valued basis functions $\phi_1(t), \phi_2(t), \dots, \phi_N(t)$ form an orthonormal set

$$\int_0^T \phi_i(t) \phi_j(t) dt = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases} \quad \text{--- (3)}$$

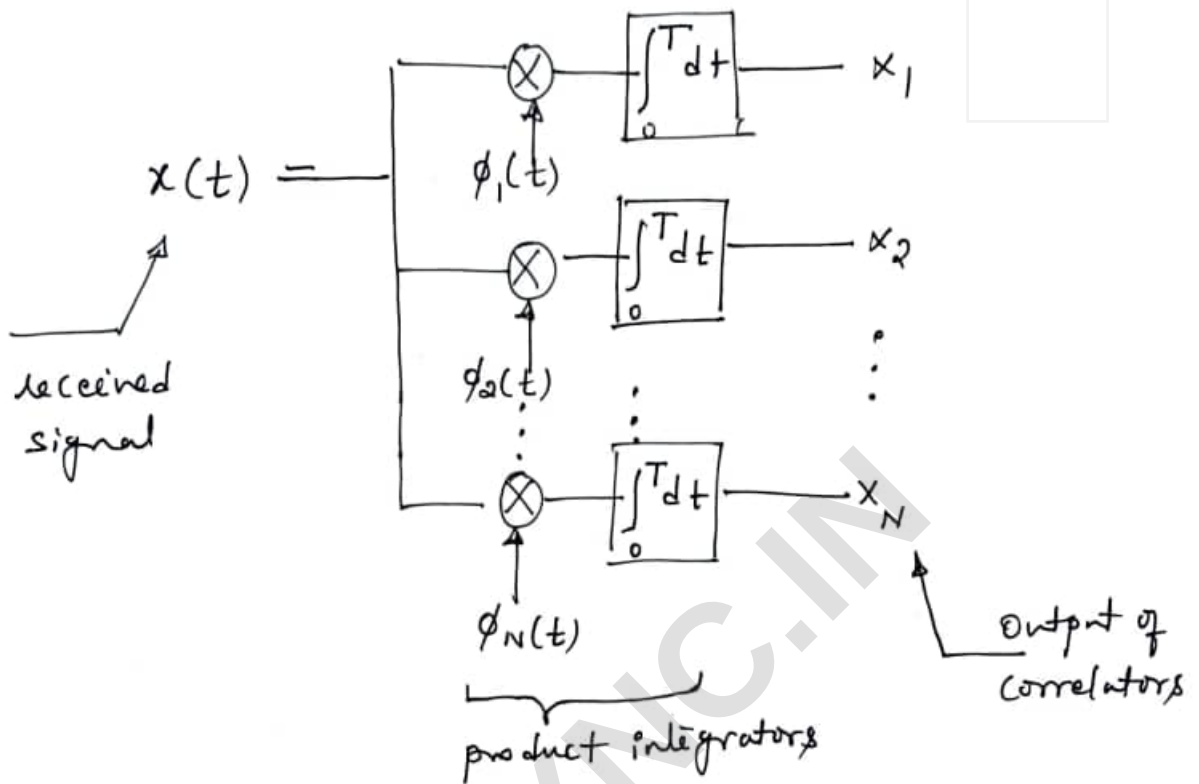
then received signal $x(t)$ is

$$x(t) = s_i(t) + n(t) \quad \text{--- (4)}$$

$$, \quad \begin{cases} 0 \leq t \leq T \\ i = 1, 2, \dots, M \end{cases}$$

where $n(t)$ is sample function of white Gaussian noise process $N(t)$ of zero mean & PSD $N_0/2$.

if input to the bank of N product integrators or correlators is received signal $x(t)$ eqn (4)



the output of correlator j , is sample value of random variable X_j , whose sample value defined by

$$x_j = \int_0^T x(t) \phi_j(t) dt \quad \text{--- (5)}$$

substitute eqn. (4) here.

$$\begin{aligned} \therefore x_j &= \int_0^T (s_i(t) + w(t)) \phi_j(t) dt \\ &= \underbrace{\int_0^T s_i(t) \phi_j(t) dt}_{\text{eqn (2)}} + \underbrace{\int_0^T w(t) \phi_j(t) dt}_{w_j} \end{aligned}$$

eqn (2)
 s_{ij} deterministic component of x_j due to transmitted signal $s_i(t)$

w_j is sample value of random variable w_j due to channel noise.

$$\therefore x_j = s_{ij} + w_j \quad \text{--- (6)}$$

Let us consider a new stochastic process $X'(t)$ whose sample function $x'(t)$ related to received signal $x(t)$

$$x'(t) = x(t) - \sum_{j=1}^N x_j \phi_j(t) \quad \text{--- (7)}$$

substitute eqn (4)

eqn (6) substitute here

$$\begin{aligned} \therefore x'(t) &= s_i(t) + w(t) - \sum_{j=1}^N (s_{ij} + w_j) \phi_j(t) \\ &= s_i(t) + w(t) - \underbrace{\sum_{j=1}^N s_{ij} \phi_j(t)}_{s_i(t)} - \sum_{j=1}^N w_j \phi_j(t) \\ &= \cancel{s_i(t)} + w(t) - \cancel{s_i(t)} - \sum_{j=1}^N w_j \phi_j(t) \\ x'(t) &= w(t) - \sum_{j=1}^N w_j \phi_j(t) \quad \text{--- (7)} \end{aligned}$$

hence $x'(t)$ depends only on noise process $w(t)$ and not on signal $s_i(t)$

$$\therefore x'(t) = w'(t) \quad \text{--- (8)}$$

from eqn (7), $x(t)$ can be written as

$$x(t) = \sum_{j=1}^N x_j \phi_j(t) + x'(t)$$

put eqn (8)

$$\therefore x(t) = \sum_{j=1}^N x_j \phi_j(t) + w'(t) \quad \text{--- (9)}$$

hence received signal can be represented by vector form as

$$x(t) = [x_1 \ x_2 \ \dots \ x_N] \begin{bmatrix} \phi_1(t) \\ \phi_2(t) \\ \vdots \\ \phi_N(t) \end{bmatrix} + w'(t) \quad \text{--- (10)}$$

Optimum receivers using coherent detection.

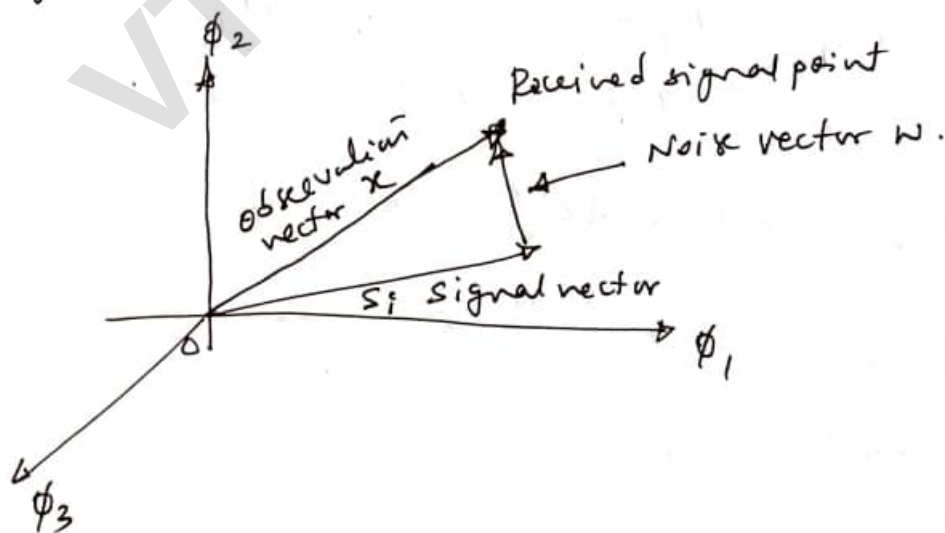
ML decoding (Maximum Likelihood decoding).

Let us consider M possible signals $s_1(t), s_2(t), \dots, s_M(t)$ transmitted with equal probability $1/M$, for T seconds.

Let $s_i(t)$ applied to bank of correlators with common input and appropriate orthonormal basis functions, yielding output vector s_i .

as $x(t)$ is corrupted by noise $w(t)$, if applied to bank of correlators, defines observation vector x .

Relationship between observation vector x and the signal vector s_i is shown below.



Signal detection statement.

"given observation vector x , perform a mapping from x to estimate $\hat{m}$ of transmitted signal m_i , in a way that would minimize probability of error in decision making process".

Let observation vector be x .

Suppose decision making is made $\hat{m} = m_i$, then average probability of error in decision is

$$P_e(m_i, x) = 1 - P(m_i \text{ sent} | x) \quad \text{--- ①}$$

to minimize the probability of error, we state here optimum decision rule.

set $\hat{m} = m_i$, if

$$P(m_i \text{ sent} | x) \geq P(m_k \text{ sent} | x) \quad \text{--- ②}$$

for all $k \neq i$, $k = 1, 2, 3, \dots, M$

This is called MAP (Maximum a posteriori probability)

for $k = i$,

set $\hat{m} = m_i$, if

$$\frac{\pi_k f_x(x | m_i)}{f_x(x)} \quad \text{--- ③}$$

Where π_k = prior probability of transmitted symbol m_k .

$f_x(x|m_i) \rightarrow$ conditional probability density function of random observation vector x given transmitted symbol m_k

$f_x(x) \rightarrow$ unconditional probability density function of x .

now, restating the rule of decision, in terms of

$L(m_k)$ as

set $\hat{m} = m_i$ if $L(m_k)$ is maximum for $k=i$

Thus, maximum likelihood decoder computes log-likelihood function as metrics for all M possible message symbols and compares them and decides in favor of maximum.

Graphical representation of decision rule.

Let Z be N dimensional space for all possible observation vector x

Let total observation space is Z partitioned into m decision region $Z_1, Z_2, \dots, Z_m$.

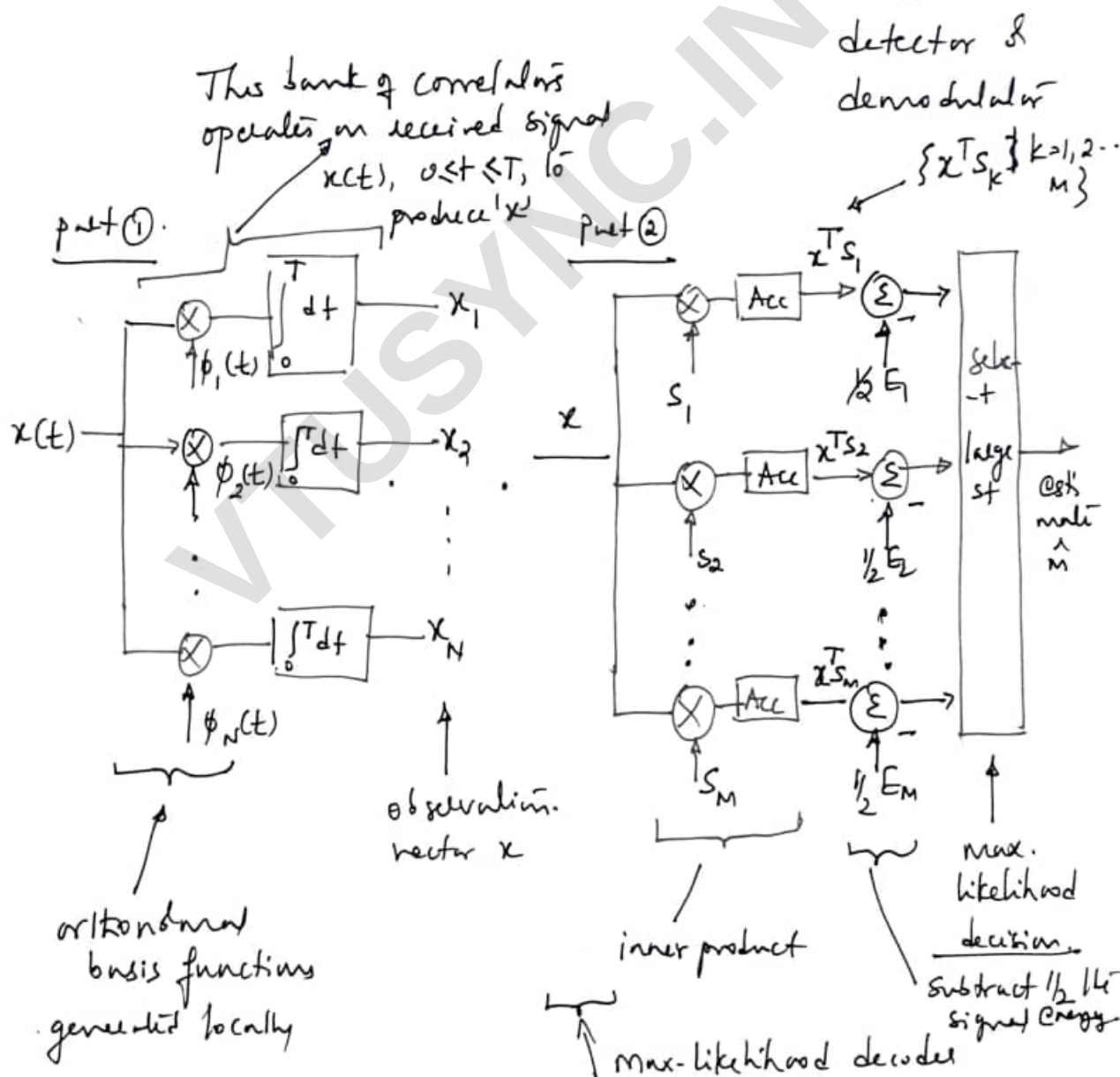
restating decision rule as

observation vector x lies in region Z_i if $L(m_k)$ is maximum for $k=i$.

Correlation Receiver:

for AWGN channel, when transmitted signals are equally likely, optimum receiver which minimizes average probability of error (P_e) is correlation receiver. which consists of two parts.

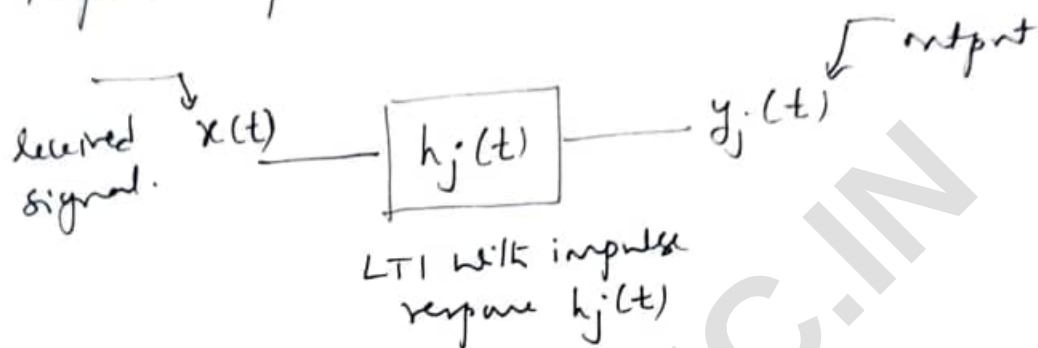
- part ① a bank of N correlators whose output is ' x '
 part ② signal transmission decoder



operate on ' x ' to produce
 nt, Professor, Dept of ECE SVIT
 (M) estimate of transmitted symbol
 (M) to reduce P_e .

Matched Filter receiver.

Correlation receiver requires analog multipliers & local reference of orthogonal functions $\phi_1(t)$ $\phi_2(t)$ $\phi_n(t)$ which are time synchronized w.r to transmitted signals. which can be avoided if, filters with appropriate impulse responses were used.



$$\therefore y_j(t) = \int_{-\infty}^{\infty} x(\tau) h_j(t-\tau) d\tau \quad \text{--- (1)}$$

integrate over duration of a transmitted signal/symbol
viz, $0 \leq t \leq T$, (replace τ by t)

$$y_j(t) = \int_{-\infty}^{\infty} x(t) h_j(T-t) dt \quad \text{--- (2)}$$

Now consider detector based on bank of correlator,
output of j^{th} correlator is given by

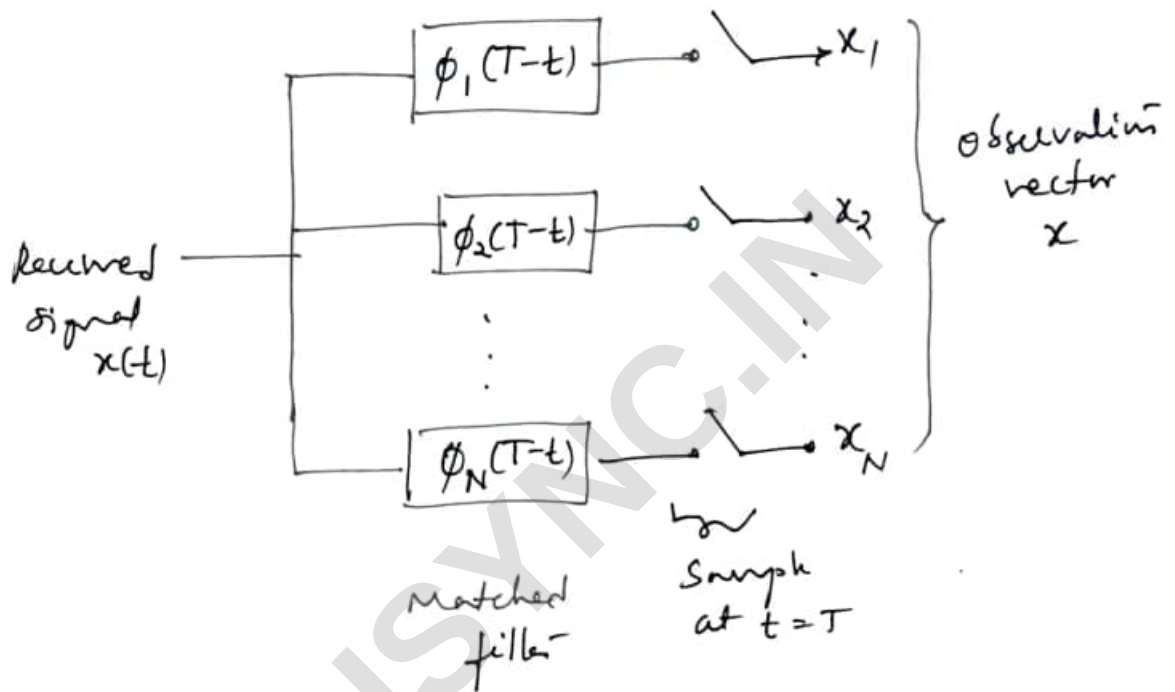
$$x_j = \int_0^T x(t) \phi_j(t) dt \quad \text{--- (3)}$$

for $y_j(t)$ to equal x_j ,

$$h_j(T-t) = \phi_j(t) \quad 0 \leq t \leq T, j = 1, 2, \dots, M$$

$\therefore$ we may express condition imposed on desired impulse response of filter as

$$h_j(t) = \phi_j(T-t) \quad 0 \leq t \leq T, j=1, 2, \dots, M.$$



$\therefore$ Given a pulse signal $\phi(t)$ occupying the interval $0 \leq t \leq T$, a LTI filter is said to be matched to the signal $\phi(t)$ if its impulse response $h(t)$ satisfies the condition

$$h(t) = \underline{\underline{\phi(T-t)}} \quad \text{for } 0 \leq t \leq T$$

A time invariant filter in this way is called matched filter. and optimum receiver using matched filter in place of correlators is called matched filter receiver.